

SOCIAL PROTECTION AND LABOR

DISCUSSION PAPER

No. 2602 | July 2025

Supporting People and Workers in an Inclusive Green Transition

Framing Social Protection and Labor
Policies to Enable and Ensure a Just
Green Transition

Maddalena Honorati
Emanuela Migliaccio

© 2025 International Bank for Reconstruction and Development / The World Bank

1818 H Street NW, Washington DC 20433

Telephone: +1 (202) 473 1000; Internet: www.worldbank.org.

This work is a product of the staff of The World Bank with external contributions. The findings, interpretations, and conclusions expressed in this work do not necessarily reflect the views of The World Bank, its Board of Executive Directors, or the governments they represent.

The World Bank does not guarantee the accuracy of the data included in this work. The boundaries, colors, denominations, and other information shown on any map in this work do not imply any judgment on the part of The World Bank concerning the legal status of any territory or the endorsement or acceptance of such boundaries.

RIGHTS AND PERMISSIONS

The material in this work is subject to copyright. Because The World Bank encourages dissemination of its knowledge, this work may be reproduced, in whole or in part, for noncommercial purposes as long as full attribution to this work is given.

Any queries on rights and licenses, including subsidiary rights, should be addressed to World Bank Publications, The World Bank Group, 1818 H Street NW, Washington, DC 20433, USA; fax: +1 (202) 522 2625; e-mail: pubrights@worldbank.org.

Supporting People and Workers in an Inclusive Green Transition

Framing Social Protection and Labor Policies to Enable and Ensure a Just Green Transition¹

Maddalena Honorati and Emanuela Migliaccio

Abstract

This paper discusses the role of social protection and labor (SPL) policies in complementing green transition policies by both mitigating the potential negative impacts of the transition on workers and low-income households and by enabling people to take advantage of emerging opportunities. The paper proposes a conceptual framework to assess how green transition policies influence people through changes in prices, production, and jobs. The paper reviews country case studies of SPL programs designed to protect people exposed to the potential negative impacts of the green transition, to prepare workers for green jobs and to promote environmentally sustainable livelihoods. Key factors of success in integrating SPL policies into green transition policies include effective institutional coordination, strong delivery systems, and integrated data systems.

Keywords: Green transition, just transition, social protection, labor, climate mitigation

JEL Codes: I38, J68, H53, Q58

¹ The authors would like to thank Trishala Kumaraswamy for research assistance and Jamele Rigolini, Sarah Coll-Black, Eliana Carranza, Alicia Marguerie, Ruslan Yemtsov, Yuko Okamura, Melis Guven, Alexander Jaeger, and Valeria Salvador for feedback at different stages of the paper's preparation and Loli Arribas Banos and Iffath Sharif for overall guidance. The paper benefited from peer review comments from Sandor Karacsony, Jeremy Lebow, Penelope Ann Mealy, Juan Jose Miranda Montero and Solene Rougeaux. The authors also gratefully acknowledge inputs and validation of country-specific examples from the following World Bank colleagues Francis Addeah Darko, Yousra Mohamed Ossama Mostafa Assaker, Timothy Joseph Peter Clay, Mawuko Abia Fumey, Tiago Falcao Silva, Joshua Gill, Yang Huang, Inchul Hwang, Lydia Kim, Elisabeth Maier, Louis Boghoyo Msuku, Federica Ricaldi, Massimo Sichinga, Julia Smolyar, Yanqin Song, Dewen Wang, and from Annabelle Bladon (IIED). Finally, the authors would like to thank all participants of the Joint Climate Change and Social Protection and Labor GSG Community of Practice, held on April 30, 2025, for their constructive feedback and suggestions.

Table of Contents

Executive Summary.....	3
1. Introduction	4
2. What do we mean by “green transition” policies?	6
3. Framing social protection and labor policies in the green transition	11
4. Social protection to ensure just green transition: practice and evidence	17
Helping poor and vulnerable households cope with price increases	17
Helping poor and vulnerable households cope with disruptions to natural-resource-based livelihoods	20
Helping workers cope with job losses and the changing task content of jobs	23
5. Social protection to enable just green transitions: practice and evidence	27
Preparing and connecting workers to green jobs.....	27
Promoting environmentally sustainable livelihoods	32
Leveraging SPL delivery systems to enhance other sectoral green transition policies	37
6. Conclusions	41
References	44
Annex 1. Overview of selected policy databases.....	55
Annex 2. OECD climate actions and policies measurement framework.....	56

Executive Summary

The green transition, characterized by a shift toward an economic model that is not based on fossil fuels and the overconsumption of natural resources, could have considerable implications for households and workers globally.

While the transition promises both environmental and economic benefits, there are several potential challenges for policy makers to address. For example, the green transition might create new job opportunities for workers in renewable energy, green construction, and sustainable agriculture, while at the same time also requiring the reskilling and upskilling of the workforce. Potential challenges include job displacements in traditional sectors, disrupted livelihoods, and unequal access to emerging economic opportunities. Additionally, consumers could encounter rising prices for goods and services that still rely on high use of fossil fuel in the production process. Such price increases may hit low-income households the hardest, with potential impacts on their purchasing behaviors and welfare.

This paper discusses the role of social protection and labor (SPL) policies in complementing green transition policies by both mitigating the potential negative impacts on workers and low-income households, and by enabling them to take advantage of emerging opportunities. To this end, the paper proposes a framework for assessing the channels through which green transition policies impact people through changes in prices, production, and labor demand. It then reviews existing evidence on the distributional impacts of green transition policies and considers which SPL policy instruments could most effectively serve as complements, based on country context.

The review finds that while all social protection instruments—within social assistance, social insurance, and labor market policies— can support green transitions, social assistance and labor market policies are more commonly used than social insurance. The relevant SPL instrument could be introduced as new initiatives or integrated into the expansion of existing programs. Either way, their success depends on effective institutional coordination (across government agencies and levels), robust data, and well-designed targeting. Granular, up-to-date data, stored in systems that are interoperable across key agencies, are necessary to identify vulnerable people, as a foundation to better target and measure the impacts of green transition policies on people. Precise targeting and compatibility with incentives to reduce the use of natural resources are particularly important for social protection measures that complement natural resource management policies. Benefits need to be carefully designed to hold back households from the poverty line while at the same time encouraging workers to stay in the labor market (and reskill or upskill as needed). Meanwhile, distributing benefits equitably across regions and people who are more at disadvantage is of paramount importance, as geographically marginalized groups, lower educated, persons with disability and women may face unique challenges to reap the benefits of the green transition.

The country examples of SPL policies and programs complementing the green transition presented in this paper highlight how social protection plays a critical role in supporting households and workers as production and consumption shift away from the overconsumption of natural resources and reliance on high carbon emissions. However, further empirical research is required to evaluate the efficacy of specific social protection initiatives in supporting a just green transition.

1. Introduction

The green transition, characterized by the shift toward sustainable and environmentally friendly production and consumption patterns, has profound implications for people and workers across the globe. As firms adapt to new regulations and technologies aimed at reducing carbon footprints, the labor market will undergo significant transformation. The green transition not only creates new job opportunities in renewable energy, green construction, and sustainable agriculture but also necessitates the reskilling and upskilling of the existing workforce. While many environmental policies aim to reduce both environmental degradation and poverty, some may have short-term unintended consequences on people and communities that depend on the protected natural resources for their livelihoods.

The green transition also affects consumers, influencing their purchasing behaviors, lifestyle choices, and overall welfare. This shift often leads to higher initial costs for greener products and services such as electric vehicles, higher electricity bills and energy-efficient appliances—costs that have a disproportional impact on low-income households. While the green transition promises environmental and economic benefits, it also poses challenges such as job displacement and disrupted livelihoods in traditional sectors. Understanding and addressing these impacts is crucial for fostering a just and inclusive transition that benefits all people, ensuring equitable access to new economic opportunities. Greater understanding also helps secure the political support needed to maintain and advance national policy objectives.

Social protection and labor (SPL) policies play a critical role in mitigating the potential negative impacts of the green transition on people, a role that is well known and understood. Equally important—though it receives less focus in the literature—is the role that SPL policies play in reskilling workers and facilitating their transition to green jobs, and supporting behavioral changes toward better environmental practices and sustainable livelihoods, partly through outreach and incentives embedded in program design.

The objective of this paper is to position SPL policies as part of an integrated policy response to challenges such as pollution, natural capital degradation, and loss of biodiversity by showing how SPL policies—and their adaptations—can serve to both *promote* the green transition and *protect* people who may be negatively affected by it. To this end, the paper proposes a framework for assessing the channels through which green transition policies impact people, reviews existing evidence on the distributional impacts of green transition policies, and offers suggestions on the types of SPL policy instruments to be deployed to address specific green transition risks while ensuring the equitable distribution of related benefits.

The paper focuses on the intersection of SPL, decarbonization, and environmental conservation policies, rather than the contribution of SPL policies to the broader climate adaptation and mitigation agenda (ILO 2024; World Bank 2024b). Building on previous work (Costella et al. 2024; Rigolini 2021; Keese and Marcolin 2023) and reflecting the challenges and social protection solutions discussed in World Bank Country Climate Development Reports (CCDRs) and a CCDR human capital synthesis note published in 2024 (World Bank 2024b), the paper provides a new conceptual framework for the role of SPL policies in the green transition, a typology of which SPL policy instruments best complement specific green transition policies—and a new focus on the implementation of these instruments and delivery of relevant services. In addition, the paper expands the evidence on the actual impacts of decarbonization and environmental policies on people, including workers, and provides illustrative country examples of innovative SPL policy solutions to support the shift to green jobs and environmentally sustainable livelihoods, to ensure energy affordability and energy-efficient consumption, and to mitigate the income and job losses due to the green

transition. The paper's target audience includes World Bank task teams, social protection practitioners, government officials, and the broader international development community committed to reducing poverty and inequality on a livable planet. It outlines policy and operational considerations for the inclusion of SPL policies in Nationally Determined Contributions (NDCs) and as part of environmentally sustainable and inclusive growth strategies. Also, the paper further expands the discussion of the strong interconnections between environmental conditions and jobs seen in the Reboot Development report (World Bank, forthcoming) by proposing a conceptual framework to organize the contribution of SPL policies to environmental goals and practical examples of policy solutions.

High-income countries have been the first to adopt green transition policies, particularly those focused on decarbonization, along with many accompanying social protection measures. Meanwhile, as a growing number of middle- and low-income countries shift away from economic models reliant on fossil fuels and the overconsumption of natural resources, it becomes increasingly important for them to adopt SPL policy instruments to prepare and connect workers to the green economy.

The paper finds that the empirical evidence evaluating the impacts of social protection policies in supporting the green transition and make it “just” is still limited, in part due to these policies' relatively recent adoption. New forms of social assistance, social insurance, and labor market programs are emerging to both protect workers and poor households from the possible adverse effects of green transition policies and to help them seize emerging opportunities. The paper finds that social assistance and labor market policies are more commonly used to promote the green transition than social insurance instruments.

Depending on the context, SPL policies supporting the green transition can be implemented either through the establishment of new initiatives or by adapting existing programs. Key elements of their successful implementation include:

- Effective institutional coordination between ministries of social affairs and labor with other sectoral ministries and agencies responsible for green transition policies and their implementation (i.e., energy, environment, agriculture);
- Streamlined implementation arrangements between central and local governments;
- SPL delivery systems that are integrated with the administrative systems of other relevant government agencies implementing green transition policies; and
- Digital and interoperable data systems.

The remaining part of the paper is structured as follows: section 2 proposes a definition of green transition and proposes a taxonomy of SPL policies complementary to selected green transition policies discussed in the paper. Section 3 presents the framework used to assess the channels through which green transition policies affect people, and the types of SPL policies needed in response while providing a review of the available evidence quantifying the impacts of the green transition on people's income, jobs, and employability. Section 4 discusses country examples of the adaptation of existing SPL programs and the introduction of new social protection programs to mitigate the distributional impacts of green transition policies and ensure inclusive transitions. Section 5 provides country examples of SPL programs to enable the green transition by preparing and connecting workers to the green economy, promoting environmentally sustainable income-generating activities, and promoting energy-efficient and environmentally friendly consumption choices. Section 6 concludes by summarizing key conclusions and highlighting remaining knowledge gaps.

2. What do we mean by “green transition” policies?

While the term “green transition” is used widely in the policy arena, there is neither a unique and internationally agreed definition, nor broad agreement as to what is meant by green transition policies. Green transition policies can be defined either broadly to include climate mitigation and adaptation policies as well as other measures to increase environmental sustainability, or more narrowly to include only climate mitigation policies. For example, the European Union (EU) uses the term “green transition” to refer to “a process towards a new development model that ensures environmentally sustainable and fairer societies. It is a necessity to address the human-induced climate change emergency, environmental degradation (water, land, forests, atmosphere) as well as the loss of biodiversity” (European Training Foundation, 2022). Similar concepts are also implied when referring to “green growth” or the “green economy.” For instance, the World Bank has used the term “green growth” to refer to “growth that is efficient in its use of natural resources, clean in that it minimizes pollution and environmental impacts, and resilient in that it accounts for natural hazards and the role of environmental management and natural capital in preventing physical disasters” (World Bank 2012).

Similarly to “green transition,” the term “just transition” is widely used but lacks a commonly agreed definition. The term originated in the US labor movement in the 1970s when unionized workers threatened by increased regulation of polluting industries argued for a just transition. Today, the concept has been reinterpreted, recognizing that the green transition can have significant impacts on workers and communities that depend on fossil fuels and the overconsumption of natural resources. A just transition means transforming the economy and economic system in a way that is as fair and inclusive as possible to everyone concerned, creating decent work opportunities, and leaving no one behind (UNFCCC 2023; ILO 2015a).

Citizens’ attitudes toward the green transition are generally positive, though there is often a view that governments need to do more to ensure that the green transition is just. For example, the European Green Deal, launched in 2019, sets out the EU strategy to become the first climate-neutral continent by 2050 (European Commission N.d.). Financing mechanisms² have been established to support EU Member States in achieving the targets and a fair transition toward climate neutrality, and in best addressing the social and labor aspects of the green transition through active support to quality employment (job search assistance, flexible learning of green and digital skills, hiring incentives for vulnerable people), equal access to education, training and lifelong learning, reducing the tax wedge for low and middle income groups, and access to affordable essential services and housing. A vast majority (88 percent) of EU citizens support the green transition, however only half of them think that their governments (national, regional, city, or local public authorities) are doing enough to ensure the transition is fair (European Commission 2022).

There are several classifications used to categorize green transition policies. Publicly available databases of environmental and green transition policy span thousands of policies—mostly in higher-income countries (annex 1). The largest online databases include the International Energy Agency (IEA) policy database on energy policies to reduce carbon emissions, the Climate Policies database, the Climate Policy

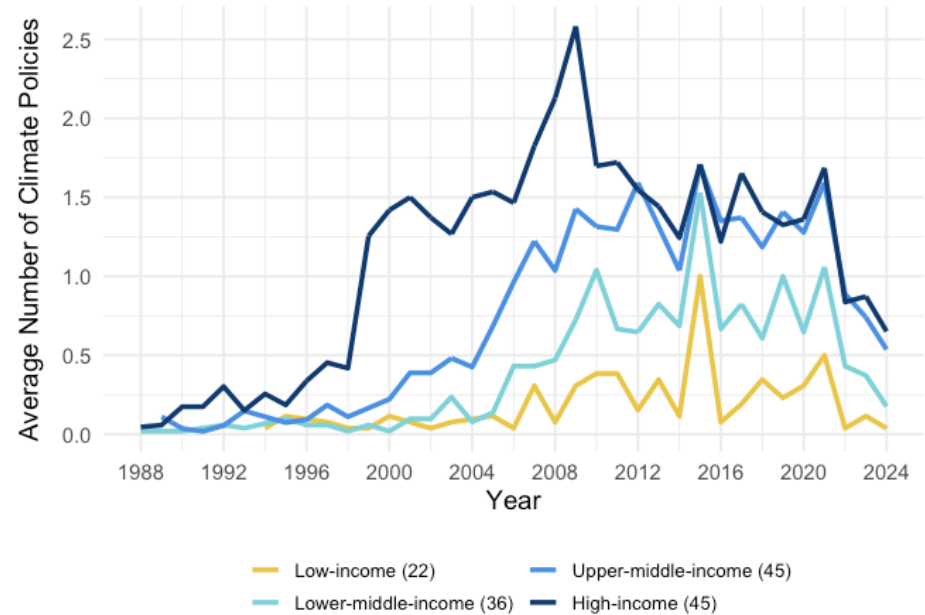
² The Recovery and Resilience Facility (RRF), the Just Transition Mechanism, the European Social Fund Plus, and the Social Climate Fund.

Radar, the Organization for Economic Co-operation and Development’s (OECD’s) Policy Instruments for the Environment (PINE), the OECD’s climate actions and policies measurement framework (CAPMF), and the World Bank’s carbon pricing database, which exclusively focuses on carbon tax policies. Each database uses a different taxonomy to classify policies. For example, the OECD CAPMF categorizes policies into eleven types within three broad categories: market based, not market based, and a third category that includes national commitments, international compliance, and policies for climate governance (annex 2).

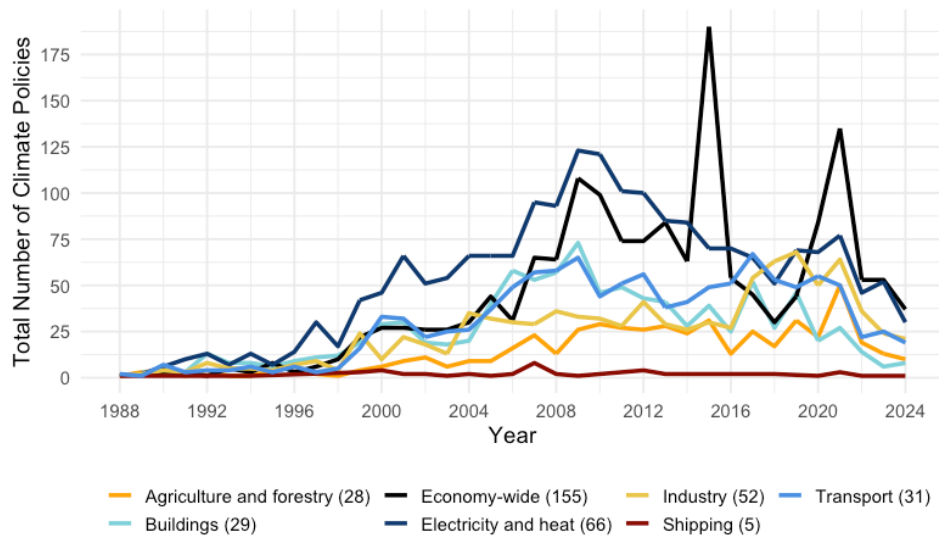
According to the Climate Policy Database, the average number of climate policies announced in the past 40 years has been much higher in high-income countries. It has to be noted that the Climate Policy Database mostly includes climate mitigation policies, just a subset of the green transition policies considered in this paper. Climate policies most commonly target electricity, transport, and construction industries. Middle- and lower-income countries, mostly in East Asia and Sub-Saharan Africa, have been increasingly adopting such policies since the early 2000s, with a focus on agriculture and forestry policies. While sector-specific climate policy announcements have slowed, economywide policy announcements remain high (figure 2.1).

Figure 2.1. Climate policy announcements

a. By country income group



b. By economic sector



Source: Climate Policy Database 2025: <https://climatepolicydatabase.org/>.

Note: The average number of policies in Figure 2a is estimated by dividing the total policy announcements count in each income group by the number of countries in each country income group according to the WB FY25 classification (25 Low-income", 50 Lower-middle-income, 54 "Upper-middle-income (54)" and 87 "High-income (87)" held constant throughout the entire period. The maximum number of countries included in each income group in figure 2.1a and in each sector in figure 2.1b are shown in parentheses. The sector classification is based on the subsector assignments in the Climate Policy Database. Policies included in this database are often a combination of policies with an explicit climate change mitigation objective, such as greenhouse gas emission reduction strategies; energy policies that help to decarbonize the energy supply and/or reduce energy demand; and policies that aim to introduce low-emission practices and technologies to nonenergy sectors, such as agriculture and land use.

In the remaining of this paper, the term green transition refers to a shift toward an economic model that is not based on fossil fuels and the overconsumption of natural resources. In light of this definition, green transition policies include: (1) policies aimed at natural resource management and environmental protection; and (2) decarbonization policies aimed at shifting energy production to cleaner sources, such as carbon taxes, closure of coal mines, subsidies for the energy efficiency of buildings, and incentives for the adoption of cleaner production technologies.³

Policies contributing to natural resource management and environmental protection aim to reduce environmental degradation and restore ecosystems. They include, for example, policies restricting access to forests through logging bans, policies restricting fishing through seasonal closures, setting marine protected areas and fishing bans to allow declining fish stocks to recover, and trade policies regulating the import and export of goods that may contribute to environment degradation, as well as policies to promote the environmental conservation and restoration such as subsidies for sustainable agriculture, incentives to preserve natural ecosystems and to ensure sustainable resource use, and trade policies favoring sustainable goods. The policy objectives across the two groups are not mutually exclusive, as each policy instrument could contribute to multiple policy objectives.

³ Climate adaptation policies are considered to the extent they are achieving and overlap with the abovementioned policies.

Table 2.1 illustrates some examples of green transition policies reviewed in the paper and a set of SPL policies that may either complement or enhance a given green objective without leaving anyone behind. The third and fourth columns in the table do not necessarily include the recommended SPL complementary policy, but rather examples found in the case studies reviewed in the paper.

Social protection can be defined as policies and programs that help individuals and societies to manage risk and volatility, protect them from poverty and inequality, and help them to access economic opportunities. Social protection aims to achieve this by increasing people’s resilience, equity, and opportunity—the three goals of social protection—by means of a range of instruments in the domains of social insurance, labor and economic inclusion, social assistance and social care (World Bank 2022b).

The following sections will review SPL policies adopted to either support an inclusive, just transition or further enhance the social and employment co-benefits of the green transition. Table 2.1 summarizes the country examples of SPL response to specific green transition policies that we reviewed in the paper as illustrative examples of the complementary role of SPL policies in the green transition. While social protection policies are increasingly recognized as fundamental to the green transition, a review of the latest round of the NDCs found that only 13 percent mention and integrate social protection policies (Crumpler et al. 2024), to support both climate adaptation and mitigation objectives. For example, the Malawi NDC recognizes the potential mitigation benefits of public work programs, and Antigua and Barbuda’s NDC is committed to training the affected national workforce in cleaner technologies, providing financial support to green entrepreneurs and businesses, and assisting farmers and their families in coping with drought and hurricanes (NDC Partnership 2025; Government of Antigua and Barbuda 2021).⁴ As countries update their NDCs in 2025, SPL efforts could be further featured as important policy instruments.⁵

Table 2.1 Types of green transition and social protection and labor policies considered in the paper

Objective of green transition policy	Example of green transition policy	Types of complementary social protection policies	Country examples
Environmental protection and natural resource management	Fisheries management measures—fishing bans and closures	Food transfers, cash transfers, public works, early retirement, unemployment insurance	Bangladesh, Philippines, Canada, Brazil, Republic of Peru
	Forest management measures—logging bans	Economic inclusion programs for alternative livelihoods, employment services, unemployment benefits	Kenya, China
	Ban on imports of chemical fertilizers	Economic inclusion programs targeted to poor farmers	Sri Lanka
	Payments for urban environmental services	Wage subsidies	Brazil
	Introduction of new technologies for green agriculture	In-kind transfers, cash transfers integrated with agricultural programs,	Malawi

⁴ In the case of Malawi, public works is recognized as a key tool for social support, and disaster risk management assessment and monitoring is planned to be integrated in all sectors, including public works (Republic of Malawi 2021).

⁵ <https://www.climatewatchdata.org/ndc-tracker>.

		training on green practices and skills	
		Vocational skills training, upskilling, and reskilling	China, Ghana
	Environmentally beneficial subsidies and payments	Conditional cash transfers	Paraguay, Brazil
		Use of social registry, payment and targeting systems	Brazil, Côte d'Ivoire
		Green public works	Ethiopia, India, Malawi, Fiji
Decarbonization (greenhouse gas emission reduction)	Closure of coal and oil shale mines	Unemployment benefits, severance payments, retraining, early retirement	Poland, Bosnia and Herzegovina, South Africa, Estonia
	Removal of fossil fuel subsidies	Cash transfers targeted to the monetary and energy poor	Indonesia, Dominican Republic, Arab Republic of Egypt, Ukraine, Romania
	Carbon taxes, fuel excise taxes	Universal and targeted cash transfers	Austria, Canada
	Subsidies for energy-efficient building renovations	Social registries to identify low-income households	Poland, United States
	Subsidies for energy-efficient heating and cooking		Mongolia, India
	Cleaner production and technology standards	Labor market information system and observatories for green jobs and skills	India, France
	Government direct investment in and financing mechanisms to lower-emitting industries	Intermediation and job matching services	Slovenia
		Skills development and training programs	India, Arab Republic of Egypt, Jordan, Iraq, Morocco, Tunisia, Kenya, Ghana, Argentina, Greece
		Global skill partnerships	India, Pacific Islands
		Support for green business start-ups	Croatia, Kenya

Source: Original compilation.

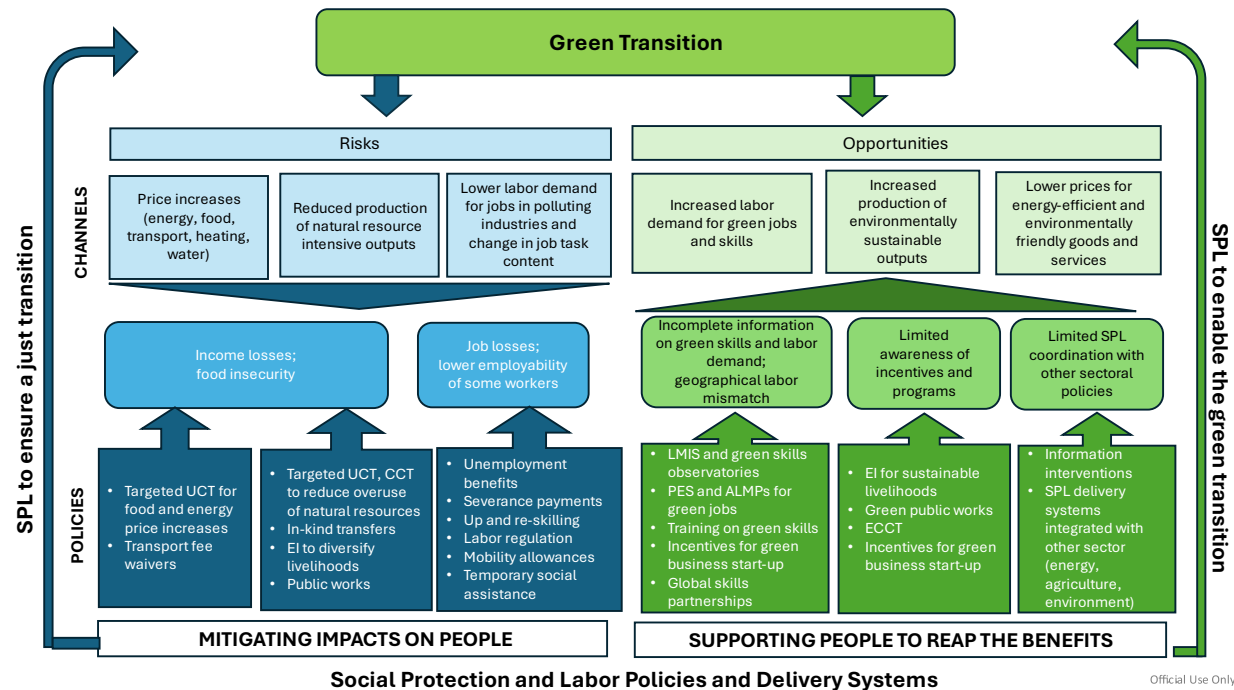
3. Framing social protection and labor policies in the green transition

The green transition poses both opportunities and risks for workers, small-holder producers and farmers, and vulnerable consumers, who are the focus of SPL policies. The risks of the green transition are associated with the increased cost of basic goods and services, job displacement in carbon-intensive and natural resource-intensive industries, and increased inequalities. In fact, the poorest and people whose livelihoods are dependent on overconsumed natural resources are most at risk of experiencing income and consumption losses due to policies restricting their production, making agriculture inputs more costly and increasing the price of basic goods and services. The green transition also stimulates new markets, particularly in renewable energy, energy efficiency, sustainable agriculture, and technologies, which in turn can lead to job creation in these emerging sectors. To help frame the complementary role of SPL policies, this paper identifies three channels through which the green transition can create both opportunities and risks for workers, producers, and consumers: through changes in the prices of basic goods and services (such as food, heating, utilities) likely to affect poorer households, changes in the production of outputs by self-employed and small-holder farmers, and changes in the demand for jobs regulated by labor markets.

Figure 3.1 visualizes the conceptual framework used in this paper to emphasize the role of SPL policies and programs in ensuring a just green transition and to enable the green transition by reinforcing the welfare, employment, and environmental impacts of green transition policies. The figure outlines green transition policies described in the previous section and, more broadly, shifts driven by technological change and market forces. The left-hand boxes (in blue) categorize SPL policies and programs to mitigate the risks and shocks that poor and vulnerable people may experience during the transition, such as food insecurity, welfare, income, and job losses. The right-hand boxes (in green) present SPL policies and programs to support people and workers in seizing the opportunities of the green transition by closing information and skills gaps to access green jobs, reducing geographical mismatch, raising information, increasing awareness of green transition policies, and facilitating people's access to subsidies and incentives for energy efficient and environmentally friendly consumption through integrated delivery systems.

While all SPL policies listed in both the left- and right-hand sides of figure 3.1 would need to be supported by solid SPL delivery systems and clear institutional arrangements, to enhance the impacts and take-up of green sectoral policies, SPL delivery and data systems (social registries, payment and targeting systems) need to be integrated with other government agencies in energy, environment, and agriculture sectors, for example. Finally, it has to be noted that some SPL instruments, such as economic inclusion and active labor market programs (ALMPs), could be used both to protect people from the risks of the green transition and to enable people to seize the opportunities of the green transition, and therefore appear on both sides of figure 3.1.

Figure 3.1 Framing Social Protection and Labor policies for the green transition



Source: Original compilation.

Note: ALMP = active labor market program; ECCT = environmental conditional cash transfer; EI = economic inclusion; LMIS = labor market information system; PES = public employment services; UCT = unconditional cash transfer.

The risks associated with the green transition are related to welfare and income losses due to the increased price of energy for household consumers, food, water and transport. The removal of fossil fuel energy subsidies are found to disproportionately affect low-income households given the larger share of their budget spent on such basic expenditures (Olivier, Matytsin, and Gencer 2023; Gencer and Arizu 2024; Mukherjee et al. 2023). The distributional effects of climate mitigation policies such as carbon pricing for electricity, heating, and maritime transport can be regressive as low-income households tend to spend larger shares of their income on these items and on imported goods. Conversely, carbon pricing for air transport can be progressive, as high-income households use air travel at higher rates. Agriculture-related policies (such as taxes and standards) may also be regressive as low-income households may spend a larger income share on food, however, this tends not to be true for high-emission foods (D'Arcangelo et al. 2022).

In low-income countries, carbon pricing may have a more neutral or progressive effect as low-income households have limited access to energy-consuming assets and services and may therefore be less exposed than high-income households to an increase in fossil fuel prices. Dorband et al. (2019) find that for countries with per capita incomes below US\$15,000 per year, carbon pricing has, on average, progressive distributional effects (Dorband et al. 2019).⁶ Among upper-middle-income countries, the

⁶ The authors also show that distributional outcomes are primarily determined by differences among income groups in consumption patterns of energy, rather than of food, goods, or services. The authors conclude by arguing that an inverse U-shape relationship between energy expenditure shares and income explains why carbon pricing tends to be regressive in countries with relatively higher income (Dorband et al. 2019).

evidence is mixed and varies with levels of access to public transport and electricity and other lower-carbon alternatives (Hallegatte et al. 2024).

Few empirical studies examine the distributional impacts of nontax regulations. For instance, the effects of fuel efficiency standards for vehicles in the United States are more regressive than energy taxes (Levinson 2019).⁷ Strict building energy codes in California reduce homes' square footage and home values for lower-income households (Bruegge, Deryugina, and Myers 2019).

Second, some environmental policies such as fishing and logging bans may have the short-term unintended consequence of reducing the production and therefore the income of self-employed people (fishers, farmers, and other small and subsistence producers) whose livelihoods depend on the utilization of the natural resources that are either banned or regulated. For example, Kenya implemented a national moratorium on logging activities in public forests,⁸ impacting the livelihood of thousands of people relying on charcoal as their main source of income and of more than a million households relying on charcoal as a primary source of fuel for cooking.⁹ The moratorium resulted in income losses in the sector, with many sawmills operating at about 35 percent of their installed capacity.¹⁰ In May 2021, Sri Lanka imposed a ban on the import of agrochemical fertilizers, which seemed to have led to a historical 32 percent decline in rice yields, with losses disproportionately affecting landowners and regions specializing in fertilizer-intensive crops. Farmers suffered losses equivalent to 4–8 percent of their baseline income, depending on their geographic location (Ghose, Fraga, and Fernandes 2023).

Agriculture subsidies are intended to provide price stability, food security, and support to farmers, but often lead to unintended inefficiencies and environmental harm. Agricultural subsidies often benefit wealthier households more than poorer ones, as they are tied to production levels, favoring larger farms (Damania et al. 2023). However, agricultural subsidies constitute a significant share of the income of poor households, implying that removing these subsidies without compensation can be harmful.

Impacts from fishery subsidy reforms on low-income households can be through changes in prices, production, or jobs, depending on the context. First, the removal of fishery subsidies may result in increased fish prices, thus affecting the capacity of low-income households to buy food and maintain nutritionally balanced meals. Second, in the case where the reforms reduce or cut subsidies for small-scale fishers, this may strongly affect their main source of livelihood. Third, when fishers work as crew on industrial fishing vessels, the removal of subsidies may lead to closure or restructuring of the fishing fleet, thus leaving crew members or processors unemployed. While fishing subsidies tend to be regressive as they mostly subsidize industrial fishing fleets, the distributional consequences of removing fishery subsidies can be difficult to assess (Damania et al. 2023).

⁷ The author shows that taxing gas would be less regressive than regulating the fuel economy of cars if the two policies are compared on a revenue-equivalent basis.

⁸ The ban also included a ban on domestic charcoal production in public forests.

⁹ In Kenya, about 1.5 million households rely on charcoal as their primary cooking fuel, with urban households depending more on charcoal than rural households. About a third of traditional charcoal stove users spend more than 5 percent of their total expenditure on fuels (Republic of Kenya, Ministry of Energy and Petroleum 2024; Dubey et al. 2019).

¹⁰ Technical report by the Kenya Forestry Research Institute. The economies and livelihoods of forest dependent towns and communities were also negatively affected, leading to increased poverty (Kagombe et al. 2020).

Finally, decarbonization policies may result in job losses in highly polluting economic activities and may require some workers to reskill themselves to remain productive, due for example, to firms' green technology adoption and compliance with performance standards. Job losses due to mine closures, for example, impact workers, their families, and communities, requiring tailored support in terms of income protection and reemployment assistance. At the same time, the structural changes in the task content of some jobs due, for example, to employer's adoption of green technologies make some workers irrelevant and may pose new challenges for some workers, especially those who are less employable given their skills and education, and those who are less ready to move to locations where labor demand is concentrated. For example, the EU regulation on carbon dioxide emission performance standards for cars and vans¹¹ is expected to have adverse employment consequences for workers in the manufacturing of traditional internal combustion engine vehicles and related sectors (Celasun et al. 2023).

The labor market impacts of some decarbonization and environmental policies are expected to be localized in certain geographical areas. For example, policies regulating the phaseout of coal mining have negligible impacts on national employment but substantial impacts on local labor markets due to the spatial concentration of coal-related jobs. While these jobs represent a small share of national employment, expected job losses may account for large shares of local employment (World Bank 2018). For example in Poland, coal-related employment represents between 20 and 50 percent of total employment in some municipalities in coal regions, often with sluggish labor markets, calling for place-based SPL policies (Christiansen et al. 2022 a,b,c). Similarly, in some countries, the labor market impacts of logging or fishing bans may be small nationwide but localized in communities and geographical areas highly dependent on regulated economic activities, therefore requiring complementary policies to diversify livelihoods and economic activities away from natural-resource-intensive activities.

Some workers face higher risks and are more vulnerable to green transition policies than others.¹² Workers in highly polluting industries are more exposed to the risk of job loss and displacement than workers employed in low-polluting industries. Evidence shows that workers in highly polluting industries represent about 4.2 percent of total employment on average in Europe and Central Asia, 5.7 percent on average in EU27 countries (but above 25 percent in Romania and Greece), 12.5 percent in 19 emerging and developing countries, and 4.6 percent in India.¹³ Workers employed in occupations that are undergoing substantial changes in task content and skills requirements are vulnerable to the green transition as they will need to upskill and reskill themselves to remain productive. These workers represent about 13 percent of total employment in Europe and Central Asia, 13 percent in South Asia, and 10 percent in Latin America and the Caribbean.¹⁴ In sum, workers' exposure to the effects of green transition policies is higher for those employed in highly polluting industries and in occupations requiring substantial retraining.

International evidence consistently shows that workers vulnerable to the green transition are more commonly lower educated, rural, and in the low and medium segments of the earning distribution,

¹¹ https://climate.ec.europa.eu/eu-action/transport/road-transport-reducing-co2-emissions-vehicles/co2-emission-performance-standards-cars-and-vans_en.

¹² See Knudsen et al. (forthcoming) for a discussion of labor market frictions that can hinder workers from transitioning out of declining sectors or into growing green industries.

¹³ Defined based on the methodology outlined in Vona et al. (2018); Mayer Gukovas, Garrote-Sanchez, and Makovec (forthcoming); Ham, Vazquez, and Yanez-Pagans (2025); Ham et al. (forthcoming); Alexander et al. (2024).

¹⁴ Mayer Gukovas, Garrote-Sanchez, and Makovec, forthcoming; Ham, Vazquez, and Yanez-Pagans 2025; and Winkle et al. 2024, based on the Green Economy Program and the US O*Net taxonomy of green occupations.

possibly exacerbating existing income and spatial inequalities (Mayer Gukovas, Garrote-Sanchez, and Makovec, forthcoming; Alexander et al. 2024; Mosomi and Cunningham 2024; Granata and Posadas 2024; Winkler et al. 2024). Workers in polluting industries and in occupations requiring upskilling have lower earnings on average than workers in green jobs. For example, polluting jobs account for 15–20 percent of jobs held by men in middle-class households in Brazil and South Africa, and about 10 percent of those in Colombia, the United Kingdom, and the United States. Finally, polluting jobs exhibit a gender imbalance, putting men more at risk, especially in higher-income countries (the gender imbalance is less pronounced in middle- and low-income countries; Alexander et al. 2024).

Workers' vulnerability to the green transition also depends on their capacity to acquire new skills, occupational mobility, and access to quality training, social protection, and employment services, among others. Not all workers in polluting industries are vulnerable in the same way: some of these affected workers may be able to transition easily to new job opportunities, whereas many others (older and lower-educated workers) may not. The quality of their foundational and technical skills will likely affect their transitions to green or neutral job opportunities, defined based on what tasks workers do. A review from Keese and Marcolin (2023) suggests a moderate skill-biased impact from climate change and related policies, with a particular emphasis on the growing importance of technical skills. In Europe and Central Asia countries, the skills gap to be closed among workers vulnerable to the green transition is estimated to be about 10 percent of the skills needed in the most similar occupation (Mayer Gukovas, Garrote-Sanchez, and Makovec, forthcoming). The estimated cost to close the skills gap among the 417,000 workers in need of retraining was estimated to be between €700 million and €2.4 billion for Western Balkan countries (World Bank 2025a). In Brazil, agricultural workers tend to be more vulnerable as they are less occupationally mobile (Berryman et al 2025).

Preferences also matter in determining viable job pathways. Within the coal mining sector, evidence shows that workers at risk of losing their jobs are reluctant to move to different places where better job opportunities are and have high reservation wages. In coal communities, miners exhibit high reservation wages compared to non-mine workers. This reflects miners' usually higher earnings, due the dangerous nature of mining work, and—in some countries—the presence of strong unions. These results are consistent across countries in Bosnia, China, and Poland (World Bank 2025a; World Bank, forthcoming; Christiansen et al. 2022).

Informal workers are more vulnerable to the risks posed by the green transition than formal workers covered by unemployment benefits and other social insurance providing income protection against the risk of job loss. In Poland, formal unionized workers directly employed by coal mine companies are entitled to additional categorical social protection benefits adopted to mitigate the impacts of mine closures; therefore, they are more protected than non-mine workers living in affected municipalities who are indirectly affected by mine closures (Christiansen et al. 2022; Honorati and Banaszczyk 2022). In terms of access to training, workers in polluting industries have low participation rates in training across OECD regions despite the higher risk of displacement (OECD 2023). While most workers in the Western Balkans are eligible for unemployment insurance, these benefits provide on average moderate to low levels of income replacement, between 25 and 50 percent of average wages, and are not always based on workers' previous wages (World Bank 2025a). SPL policies that help people and workers cope with the risks of the green transition require careful planning, combining different policy responses (targeted cash transfers, upskilling and reskilling, unemployment insurance, temporary income support and mobility allowances,

and housing support, among others) and solutions tailored to workers' characteristics (such as age, skills, and readiness to move, be trained, or reemployed in new sectors and occupations) and to households' income, consumption, and food security characteristics, while at the same time providing the right incentives to contribute to environmental and employment outcomes.

Green transition is also bringing opportunities to people. The opportunities are associated with monetary incentives for employers producing environmentally sustainable outputs and adopting clean technologies, lower prices for consumers of energy-efficient and environmentally friendly goods and services, and the increased demand for workers in green industries and occupations.¹⁵ Positive labor market impacts are observed in industries more likely to benefit from green investments and where the job creation potential is highest, for example, in the renewable energy and energy efficiency sectors (IRENA and ILO 2023), and for some occupations in other sectors relevant to decarbonization efforts (such as carpenters and welders, required in the construction of new energy-efficient buildings, and agricultural workers needed for organic farming).¹⁶ When helping people seize opportunities from the green transition, SPL programs can also channel climate finance to vulnerable people and workers.

A recent review of 60 papers evaluating the employment effects of climate mitigation policies indicates that such impacts are likely to be modest, either net positive or neutral, and that indirect or induced employment is a significant driver of positive outcomes, underscoring the necessity of designing policies that maximize and sustain these effects (Godinho 2022). Similarly, Keese and Marcolin (2023) review the literature and find that past climate change mitigation policies have not led to significant reductions in aggregate employment in manufacturing industries, though effects vary across sectors and country contexts. Globally, structural economic models of transitions to low-carbon economies predict modest net job growth, concentrated in certain sectors and locations (ILO 2018; IEA 2022a; Hafstead and Williams 2020; Alexandri, Anton, and Lewney 2024). Green transition policies have unequal impacts across regions. For example, subsidies for renewable energy production are likely to benefit regions with natural endowments (wind and sun) and developed energy-efficient infrastructure.

While the green transition is widely argued as having the potential to create jobs, if the workforce is not ready and in the right place, these jobs will remain unfilled, delaying the decarbonization process and reducing the payoffs of other public investments. To make the green transition happen, it is important to equip the workforce with the skills needed to reduce the environmental footprint of production processes. To meet the increased demand for green jobs, upskilling and reskilling the workforce, ALMPs, and public employment services targeting green jobs and short-term training in green skills for the unemployed, job seekers, and recent graduates represent key SPL policies, while economic inclusion programs, green public works and cash transfers with conditionalities tied to environmental objectives play a key role to promote sustainable livelihoods among the low-income and marginalized households.

The reliance on social protection delivery systems to encourage compliance with and adoption of green transition policies implemented by sectors has been less explored and holds potential to be expanded. For example, incentives for green agriculture, subsidies for the increased energy efficiency of residential heating and appliances, incentives for afforestation, and biodiversity protection could be further enhanced by reaching out to poor, marginalized, and vulnerable population subgroups. Countries with well-

¹⁵ Cunningham et al. (forthcoming) for a review of definitions of green jobs.

¹⁶ US O*Net Green Economy Program.

established delivery systems with good coverage could leverage their social registries, targeting mechanisms and payment systems to increase the take-up and adoption of green transition policies by establishing interoperable data agreements, interministerial institutional arrangements, and procedures to integrate social protection with energy, environment, agriculture, and climate programs.

4. Social protection to ensure just green transition: practice and evidence

Different SPL instruments (social assistance, social insurance, and labor market programs) can be deployed to mitigate the potential negative impacts of green transition policies on poor people and affected workers. The right choice of instrument depends on the channel (price, livelihood, or job) through which people are affected, the readiness of the existing social protection system to respond to the challenges posed by the green transition, the institutional and implementation framework, and available financing mechanisms.

This section is meant to provide some illustrative examples of how SPL programs and policies have been either adapted or introduced at the national or local level to make the green transition just and fair for all. Following the conceptual framework in figure 3.1 (and specifically the left-hand section in blue), the subsections below review SPL responses to food and energy price increases due to green transition policies, the disruption of economic activities dependent on natural resources, job losses in highly polluting sectors, and changes in the task content of jobs due to the adoption of cleaner technologies, practices, and regulations.

Helping poor and vulnerable households cope with price increases

General energy subsidies, such as universal price subsidies for fossil fuels, tend to be regressive (Arze del Granado, Coady, and Gillingham 2010; Mukherjee et al. 2023). The removal of fossil fuel subsidies often lead to higher energy prices for residential consumers. In such cases, evidence shows that energy subsidy reforms have disproportionately larger impacts on low-income households, who must allocate a larger share of their overall expenditure to energy (Olivier, Matytsin, and Gencer 2023). The distributional impacts of removing fossil fuel subsidies depend on several factors, such as the specific types of energy subsidized, the consumption patterns of different income groups, and whether any compensatory measures are implemented. Effects on households are both direct, through increases in the prices of gas and electricity, and indirect, through changes in the prices of goods and services that use fuels as inputs (Damania et al. 2023). In many developing countries, increasing block tariffs or lifeline tariffs are used to cross-subsidize residents who consume less, for example, less than a social threshold. However, the use of these measures to support the poor can be problematic because the quantity of energy consumed is poorly correlated with poverty and other social vulnerabilities (Yemtsov and Moubarak 2018).

Targeted cash transfers and subsidies

To mitigate the negative impacts of fossil fuel subsidy reforms on poor and vulnerable households, targeted cash transfers have been increasingly adopted as part of energy subsidy reforms in several countries (Mukherjee et al. 2023). Targeted cash transfers have been deployed either by reforming existing social assistance cash transfer programs to provide top-up benefits (larger amounts) or to expand the coverage to new beneficiaries, or by introducing new benefits to address energy poverty, depending on the administrative capacity and coverage of the SPL system.

Critical considerations for using existing cash transfer programs as instruments to mitigate the negative impacts of price changes on welfare is their ability to cover a large share of the poor and vulnerable and the flexibility of their delivery systems in providing an increased volume of support when needed. In Ukraine, for example, the government increased spending for the Housing Utility Subsidy (HUS) program, which provides social support for the utility payments of low-income households, when it decided to eliminate gas subsidies in 2016, which caused residential electricity prices to double. The coverage of HUS was increased to 36 percent of households. This required changing the rules of the existing program (to improve targeting, eligibility checks, and oversight).

Topping up the benefit amounts of existing social assistance benefits is a good option in countries where existing programs offer high coverage of the poor but low benefit levels. In Romania, heating benefits were increased to cushion several rounds of energy reform between 1997 and 2007 (World Bank 2011).

When existing social assistance programs do not have sufficient coverage of the poor and their delivery systems are unable to support such an expansion, the introduction of new programs that address the risk of energy poverty during energy subsidy reforms is the preferred option. For example, in 2005 Indonesia introduced a temporary poverty-targeted cash transfer—the Subsidi Langsung Tunai—to accompany energy subsidy reforms. In response to the later waves of energy reforms in 2014, the government instead opted to expand existing benefits (BLSM cash transfer program, conditional cash transfer, rice subsidies, and education subsidies). In Egypt, the Ministry of Social Solidarity (MoSS) established a new cash transfer program in 2014—Takaful and Karama (“Solidarity and Dignity”)—targeted to the poor (through proxy means testing) including to mitigate the impacts of price increases on the poor. The program considerably expanded between 2013 and 2019 when energy subsidies were continuously being reduced.¹⁷ As of 2023, 3.11 million poor households were enrolled in the program, which provided compensation for income losses due to residential energy price increases (World Bank 2023b).

The Dominican Republic combined the introduction of a targeted gas subsidy with the expansion of an existing cash transfer program. The universal subsidies on gas and electricity were replaced with a targeted gas subsidy—the Bonoluz—based on the national targeting system and through the expansion of the Solidaridad conditional cash transfer program both vertically and horizontally; it provided increased benefits to existing poor beneficiaries and enrolled additional households affected by energy subsidy reforms (Costella et al. 2024).

¹⁷ Energy subsidies declined from 6.9 percent of gross domestic product (GDP) in 2013 to 1.9 percent in 2019, accompanied by a 60 percent increase in the budget allocation to existing social assistance programs to compensate poor households.

Although the above examples pertain to subsidy reforms motivated by fiscal crises, they are useful in understanding how social protection, and in particular cash transfers, can cushion poor people against energy prices in the context of eliminating fossil fuel subsidies as part of the green transition targets.

Carbon pricing is a key policy method for cutting greenhouse gas emissions by placing a financial cost on the carbon dioxide emitted from burning fossil fuels and other sources. This can be done through a carbon tax or emissions trading system (ETS), incentivizing businesses and individuals to reduce their carbon footprint. The rest of the section will focus only on carbon taxes and on the derived increased price of energy, essential goods, and services for consumers, which represent a larger share of income among poorer households (World Bank 2021a).

Microsimulation analysis shows that cash transfers mitigate the expenditure losses resulting from carbon pricing among poor households. The introduction of carbon pricing would disproportionately disadvantage households in lower income quintiles based on the tax-benefit microsimulation model applied to six countries (Ecuador, Indonesia, South Africa, Tanzania, Vietnam, and Zambia). When the revenues from carbon pricing are recycled to provide cash transfers to poor families, the overall poverty rate declines, especially if cash transfers are given as a top-up to existing cash transfer beneficiaries. The analysis simulates three types of social protection responses: increasing the benefit amount (vertical expansion), increasing the coverage of cash transfer programs (horizontal expansion), and leaving existing schemes unchanged while introducing a new unconditional scheme for selected vulnerable groups. The three types of reforms are shown to cushion the overall drop in mean income and, in four of the countries studied, result in an overall reduction in the poverty headcount from the baseline.¹⁸ Vertical expansion leads to a more significant reduction in poverty than increasing coverage in Ecuador, Indonesia, Vietnam, and South Africa. Horizontal expansion can yield better results for lower-income population groups, that is, the lower quintiles in the income distribution, in places where significant coverage gaps remain, such as Zambia (Gasior et al. 2024).

A carbon tax was introduced in 2008 in the Canadian province of British Columbia and incrementally raised until 2022 and recently cancelled in April 2025.¹⁹ A major factor contributing to political acceptance of the carbon tax was clear communication that it would be revenue-neutral, substituting other existing distortionary taxes. In 2017, the province of British Columbia redistributed 35 percent of the revenues to poor individuals (through tax cuts for the two lowest personal income groups) and 65 percent to businesses (through general corporate income tax cuts) (World Bank 2023b). While the carbon tax was found to have a positive annual 0.75 percent increase on employment (Yamazaki 2017), negative impacts were observed among lower educated male workers (Yip 2018). In particular, the carbon tax combined with the tax cuts had a negative impact on large emission-intensive firms and a positive impact on small

¹⁸ This happens in Ecuador, Indonesia, South Africa, and Vietnam but not in Tanzania and Zambia as the revenue generated is too small to fill the coverage gaps in the current social protection system.

¹⁹ In 2018, the Government of Canada enacted the Greenhouse Gas Pollution Pricing Act, mandating that all provinces and territories implement a carbon pricing system that meets a federally set minimum price per ton of carbon dioxide equivalent and establishes emissions caps under the Act. Under the Act, a federally administered backstop system applies in provinces or territories that either lack a compliant carbon pricing system or formally request the implementation of the federal system. The federal government removed the fuel charge, effective April 1, 2025, via regulations. It will also be considering broader amendments to the *Greenhouse Gas Pollution Pricing Act*, including proposed amendments to complete the orderly wind-down of the fuel charge (Government of Canada 2025).

businesses in the service sector and food/clothing manufacturing sectors (Azevedo, Wolff, and Yamazaki 2023).

Austria is implementing a national emissions pricing system and redistributes the revenues through the Klimabonus, a flat lump-sum payment for all Austrian residents and a regionally differentiated top-up based on the density of public transportation. The lump-sum payout implies that low-income households receive more than they pay via carbon pricing.²⁰ In 2024, the base amount was €145 and the top-up ranged between €50 and €145 depending on the place of residence, with higher payments for people living in areas with less access to public transport.²¹ The application-free process disbursing the lump sum to all residents required strong collaboration across various federal ministries and the merging of diverse data sets, as some Austrian residents did not have bank account information and received the bonus via postal vouchers. This process highlighted the need for up-to-date public databases, prompting improvements and data gap identification.²²

Helping poor and vulnerable households cope with disruptions to natural-resource-based livelihoods

Cash and in-kind transfers, public works, and economic inclusion programs can support households dependent on natural-resource-based livelihoods when their access to natural resources is restricted by environmental policies. For example, in-kind transfers and environmental cash for work could help poor fishers cope with reduced income and food insecurity during fishing closure seasons, while economic inclusion programs could help them diversify toward more sustainable livelihoods. These SPL interventions complement other measures often designed by ministries of environment and tourism, and implemented together with local governments, to support poor and vulnerable households who depend on the protected natural resources. These include, for instance, the redistribution of tourism revenues from protected areas and Payments for Environmental Services (Box 5.1).

Income and employment support for households and workers affected by logging and fishing restrictions

Temporary in-kind rice support has been provided by the Government of Bangladesh to assist poor hilsa fisher households during periods of restricted fishing activity. Hilsa constitutes the largest single species fished in Bangladesh, and hilsa production employs half a million poor fishers directly and another 2–2.5 million people indirectly (Islam, Mohammed, and Ali 2014). Following overexploitation in the late 1990s and early 2000s, hilsa stocks began to decline, prompting the government to implement the Hilsa Fisheries Management Action Plan (HFMAP) in 2005, which set seasonal fishing bans to protect the fish during breeding periods. To compensate for the income losses due to the ban, and reduce the risk of food insecurity among poor families dependent on fishing, the Government of Bangladesh temporarily provided rice during the ban period through the existing Vulnerable Group Feeding program targeting

²⁰ The first payments were made in August 2022; every adult received €500 and every child €250.

²¹ <https://www.klimabonus.gv.at/en/#a1>

²² EIPA Klimabonus website: <https://www.eipa.eu/epsa/klimabonus-climate-bonus/#:~:text=The%20Klimabonus%20aims%20to%20incentivise,in%20a%20given%20calendar%20year.>

vulnerable hilsa fisher households (Anwar and Cho 2019).²³ In 2019, nearly 396,000 hilsa fisher households benefited from the rice compensation (May, Velia, and Benammour 2023). In addition, the Ministry of Fisheries and Livestock introduced the Enhanced Coastal Fisheries in Bangladesh (ECOFISH-BD) project between 2014 and 2021 to enhance the capacity of communities to cope with the fishing ban.²⁴

While the fishing ban had overall positive environmental and welfare impacts in the medium term, the short-term compensation to poor households could have been improved. For instance, by design, the rice compensation scheme only targeted the most vulnerable hilsa fishers, but all fishing was prohibited by the ban in some areas, therefore leaving some affected households unprotected. Qualitative evidence indicates that the quantity of rice was not sufficient to meet household needs, as the rice distribution did not take household size into account, possibly penalizing larger households (Islam, Mohammed, and Ali 2016; Bladon et al. 2018). Further, it was found that the increased demand for rice prompted by the rice distribution to households was correlated with declining rice prices, thus adversely affecting local rice producers (Islam, Mohammed, and Ali 2014, 2016). Overall, in the medium term, the fishing bans were effective in restoring the fish stock; they allowed many fishers to recover their livelihoods and even increase profits, pointing to positive economic impacts on affected local communities and potentially positive environmental impacts.²⁵

In the Philippines, the existing public works program was expanded to support fishers affected by a temporary fishing ban in Balayan Bay in 2014. The public works program adaptation was implemented by local governments (the Provincial Government Environment and Natural Resources Office in cooperation with municipalities and barangays) and included the extension of the maximum duration of the program from 11 to 22 days, to cover the entire period of the fishing ban, and a new targeting mechanism to reach out to, identify, and select affected fishers (Altenburg et al. 2017).²⁶ Qualitative evidence indicates that the wage amount was insufficient to protect affected fishers, however, beneficiaries could see the positive environmental benefits of the temporary fishing ban (an increase in juvenile fish after 22 days) and expressed willingness to comply with other temporary bans (Altenburg et al. 2017).

To support workers affected by a fishing ban, the Canadian federal and provincial governments introduced a series of measures providing temporary income support, training allowances, and early retirement schemes (Government of Newfoundland and Labrador, n.d.). In 1992, the Canadian Federal Minister of Fisheries and Oceans declared a moratorium on the Northern Cod Fishery (which was lifted in 2024). The ban affected hundreds of coastal communities and left about 30,000 people unemployed in a province

²³ Initially 10 kilograms (kg) of rice were distributed per household; this amount was increased to 30 kg in 2009, and to 40 kg per household per month in 2013. The rice was provided for four months, to cover the ban period and one month before and after it.

²⁴ The project, implemented by the Department of Fisheries and local communities, and supported by the WorldFish Center, included various interventions to improve compliance with the ban as well as livelihood diversification strategies, including skills training, access to productive assets, and alternative income-generating activities like vegetable farming, aquaculture, and livestock rearing (May, Velia, and Benammour 2023).

²⁵ The fisheries management interventions helped the hilsa stock to recover and thus increased hilsa production in the medium term, leading to the total household fishers' income to increase by 65 percent (Haque 2020).

²⁶ The program finally covered 1,484 beneficiaries. All crew members of commercial fishing vessels were compensated for their income loss by receiving 75 percent of the regional minimum wage to perform public works. Activities included waste sorting, the cleanup of beaches, coastal areas, canals, and public markets, and similar activities (Altenburg et al. 2017).

with a population of approximately 573,000, representing about 12 percent of the province's labor force (Conway 2022; Higgins 2008). The Fishery Closure Training Allowance, instituted in 1993, provided training and income support to fishers between the time the fishery closed and when their unemployment insurance benefits started,²⁷ while the Atlantic Fishers Early Retirement Program and the Fishplant Older Workers Adjustment Program were subsequently established to support fishery workers aged 55–64 as of April 1996.²⁸ Anecdotal evidence indicates persistent high unemployment rates leading to continued dependence on government assistance and increased out-migration. While some affected workers found jobs in the emerging shellfish industry and in the tourism sector, their individual earnings remained low and often had to be supplemented with employment insurance payments, indicating the need to situate a social protection response within broader efforts to promote local economic development, in areas where green transition policies resulted in the closure of a main source of employment (Higgins 2008).

To reduce fishing of anchovetas the government of Peru introduced a vessel quota system in 2008 and promoted the creation of a special fund the transitions of workers to other sectors. The Compensation Fund for Fisheries Management (FONCOPEs) was created as an independent fund, financed by mandatory contributions from industrial fishing companies, to support affected workers through retraining, incentives to the development and promotion of small and medium enterprises (SME), and early retirement for workers aged 50-55. In its first three years FONCOPEs supported the voluntary exit of 350 workers and helped 400 workers transition to other sectors. In 2014, over 2,200 workers had been contacted, with more than 1,300 joining retraining programs and over 300 participating in SME development—mainly vessel owners and captains. However, the improved stock levels of fish led some small-scale and artisanal fishers—operating outside the vessel quota system—to sell anchoveta illegally for fishmeal, underscoring the need for inclusive fishery management across all subsectors (ILO 2015b; Bladon et al. 2022).

A combination of social insurance and employment services contributed to mitigate the income losses among workers affected by a ban introduced in China in 1998 for logging activities in natural forests in the Yangtze and Yellow River basins. The government introduced the Forest Conservation Program to provide job placement services, early retirement, and unemployment benefits targeted to about 700,000 affected workers. Recruitment offices in forest companies also helped workers transition to jobs to the newly created Forest Protection Units but also to other sectors such as tourism, construction, or transport. Four years after the beginning of the ban, about two-thirds of the workers had found new jobs or retired (ILO 2022).

In Brazil, the unemployment insurance has been amended to introduce a special scheme to insure fishers against the risk of unemployment due to temporary fishing bans, recognizing the involuntary nature of the temporary unemployment during the closed fishing season. The Unemployment Insurance for Artisanal Fishers (Seguro-Desemprego do Pescador Artesanal), also known as Seguro-Defeso, provides professional small-scale fishers who have paid some social security contributions with compensations equivalent to the

²⁷ A total of CAD2.3 million was paid to approximately 1,000 fishers.

²⁸ A chronological description of the programs is available at <https://www.gov.nl.ca/publicat/tags/text/page34.htm>.

official minimum wage for each month of fishing closure (up to five months a year).²⁹ An impact evaluation of the program finds that it benefited fishers' households (FAO 2022).

Helping workers cope with job losses and the changing task content of jobs

Coal-related jobs are at the forefront of the labor market disruptions brought by the green transition. In EU countries, more than 140,000 jobs are expected to disappear by 2030 (mostly in Poland, Germany, Romania, Bulgaria, and Greece) due to the decommissioning of coal-fired power plants and the closure of coal mines as part of the transition to a cleaner energy economy and the European Green Deal.³⁰ Coal-related jobs, including those in the mine value chain, are highly geographically concentrated around large mining conglomerates and their subcontractors, therefore requiring place-based SPL policies to mitigate the social and employment impacts of expected job losses (World Bank 2018, Ruppert et al 2021). Other traditional energy sectors relying on the production and use of fossil fuels are expected to shrink, with concomitant negative impacts on employment in these sectors.

A mix of targeted income support and employment support services—in addition to existing SPL benefits—have been adopted to support coal workers to either transition to new jobs or (for older workers) to bridge to the retirement age. However, balancing the adequacy of income support with the right incentives to work, and determining eligibility criteria that are inclusive of both directly and indirectly affected workers prove to be challenging in practice. For example, Poland introduced the Mining Social Package between 1998 and 2002 to provide income support benefits to about 67,000 mining workers through redundancy payments, social insurance, and social assistance (equivalent to about 0.75 percent of Poland's GDP at the time). There were concerns that the package was too generous, as it led about half of beneficiaries to drop out of the labor force (Ruppert Bulmer et al. 2021).

More recently, the Social Agreement the government signed with miners' trade unions in 2022/23 introduced social protection benefits for hard and lignite coal mine workers, in addition to the unemployment benefits, severance payment, public employment services, pensions, and other social insurance they are entitled to by the existing social protection legislation and the Employment Protection Law.³¹ According to the Social Agreement, miners have the right to relocate from closed mines to those still in operation, receive an early retirement package equal to 80 percent of their salaries, or receive a one-time severance payment in case of moving to a different sector of employment (European Commission 2023b). However, only workers directly employed by the mines are eligible to receive the Social Agreement benefits once mines are liquidated, leaving out affected workers in contractor companies in the mine value chain.

²⁹ The 2015 amendment to Law No. 10.779/03 explicitly defined the artisanal fisher eligible for Seguro-Defeso as a particular type of contributor, the *segurado especial* ("specially insured").

³⁰ Euracol, Joint Research Center, European Commission.

³¹ According to the employment protection law, miners would be entitled to severance payments, unemployment benefits upon registration with the local Labor Office and outplacement services as agreed with the employer and trade unions (Honorati and Banaszczyk 2022).

Similarly, the existing social protection and labor system in Bosnia and Herzegovina provides income protection to affected workers *de jure*. Mine and power plant workers are entitled to one-off severance payments for involuntary retrenchments based on tenure, protection for workers who lose work abilities due to occupational injuries³²—which are common in coal mining—unemployment insurance, public employment services, and means-tested social assistance. However, public employment services often offer low-intensity counseling activities, eligibility for last-resort social assistance is limited to households with no able-bodied members, and mine companies have social arrears and liquidity constraints that delay the payments of social benefits. Because of capacity shortcomings, *de facto* the SPL system requires adaptations to ensure a just transition for affected people, for example, through the introduction of top-up cash benefits and targeted employment services and retraining for coal miners as supported through a World Bank project (World Bank 2025a).

Upskilling and job search assistance programs to facilitate the transition of affected workers to employment opportunities in other sectors are implemented in Poland at the local level by local government units with financing support from the European Commission’s Just Transition Fund. Local labor offices in partnership with local stakeholders and the mines have intensified the provision of active labor market programs—including entrepreneurship support grants and training—in municipalities that are heavily dependent on coal production. Different from the social insurance benefits in the Social Agreement, public employment services and skills training tailored to the needs of miners are provided to all affected workers in coal regions.

For example, the Jobs after Coal program was launched in 2024 in the Eastern Greater Poland region to enhance the reemployment opportunities of 2,200 lignite mine workers and their families (Honorati and Banaszczyk 2022). The program is implemented by the Regional Labor Office in partnership with the employer (the mine) and trade unions, private employment agencies, the Agency for Enterprise Development, and the Association for Social Cooperatives. The program offers two types of employment support: one consisting of job search assistance, job matching services, and an employer subsidy per employee to support labor costs, skills training, and equipment, and a commitment to retain the new hire for at least two years. The second arm consists of business training and a grant of up to US\$3,100 per person to start a new business, preferentially in the green economy. The eligibility criteria for access to personalized employment services initially included being a former or current mine employee (or their family member). A year after the program’s start, eligibility was extended to unemployed people indirectly affected by mine closure based on geographical targeting criteria.

The Shanxi province of China employed nearly 1 million workers in the coal mining and washing industry in 2018, making up 30 percent of the country’s total workforce in this sector. Before the announcement of the “dual carbon targets” in 2020, the provincial 13th Five-Year Plan (2016–20) mandated Shanxi to phase out outdated coal capacity and modernize its energy sector. Between 2016 and 2019, 65,310 workers were displaced, most of whom were older, male, and possessed limited education. Most of them (84 percent) were supported through reassignment measures, while about 11 percent were provided with early retirement income support and a few terminated their contracts.

³² Termination of such workers requires prior consent from the miner’s trade union, and if terminated, they receive severance payments 50 percent higher than regulated by the collective agreement.

The province of Shanxi adopted additional targeted measures to support the reassignment and retirement of workers in alignment with national guidelines.³³ The reassignment of workers was supported through subsidies to coal mine companies for each worker internally reassigned to projects aimed at industrial transformation or externally reassigned to projects in newly established businesses in different economic sectors as well as through loans, incubator subsidies, and workspace for worker-led start-ups. These subsidies were in addition to the “regular” public employment services and subsidized community works directed to employees hard to reassign. Other targeted measures for miners include living allowances and continued social insurance coverage for workers within five years of retirement age.

Despite the introduction of these additional social protection and labor benefits, displaced workers faced difficulties transitioning to jobs in new sectors due to their specialized but outdated skills and the province’s weak job creation capacity in green and noncoal industries. The number of new jobs created in renewable and noncoal sectors was lower than the number of displaced workers in need of external reassignment, and the skills gap was too big to fill, based on qualitative evidence. While extensive training was provided, many workers find it difficult to adapt to new roles. On the other side, the mindset of displaced workers was more geared toward internal reassignment options—often not viable, given the limited diversification of mine companies—than external job offers, also reflecting their preference to continue to work for a state-owned enterprise (SOE) (World Bank 2025b).

In South Africa, the transition out of high-emitting sectors (coal, fuel, and heavy industries) is expected to result in about 300,000 job losses between 2022 and 2050 (World Bank 2022c). The region of Mpumalanga will be particularly affected by the closure of coal mines and coal-fired power plants. The Just Energy Transition (JET) Partnership and its Implementation Plan for 2023–27 identify several key skills interventions such as establishing a three-tier skills ecosystem to coordinate skills activities across institutions and stakeholders, creating Skills Development Zones focused on renewable energy in Mpumalanga, new energy vehicles (NEVs) in the Eastern Cape, and green hydrogen (GH2) in the Northern Cape, conducting JET skills needs assessments for these core value chains and their implications for tertiary education institutions, developing JET capacity for government and key institutions, and supporting foundational skills development by upskilling teachers and integrating sustainable energy concepts into school curricula (The Presidency Republic of South Africa 2023). The implementation started in 2024, and as of today skills development initiatives are being mapped to identify opportunities to scale up through better collaboration between government institutions and businesses (The Presidency Republic of South Africa 2025).

Targeted training programs have played a central role in facilitating a just transition in high-income countries such as in Germany and Greece. In Germany’s Ruhr region, for instance, vocational training was instrumental in supporting the shift away from coal. Specialized agencies, such as the Ruhr Coal Vocational Training Society, focus on upskilling and retraining former coal workers to help them access new employment opportunities (Galgóczy 2019). Similarly, in Greece, vocational training and reskilling form a key component of the strategy for transitioning lignite-dependent regions (IEA 2022b).

³³ China’s Ministry of Human Resources and Social Security and six other ministries issued guidelines to ensure proper displacement of workers affected by reductions in coal overproduction, and the Province of Shanxi issued their plan (Shanxi Provincial Government Office Document No. 111 [2016]).

The oil shale industry is also expected to shrink as part of the green transition, leading to similar—albeit smaller—local labor market impacts. For example, Estonia has committed to cease electricity production from oil shale by 2035. To incentivize and facilitate the transition of oil shale workers to less carbon-intensive industries, the Ministry of Social Affairs has introduced wage top-ups to compensate shale oil workers for the difference with the lower wages in less polluting jobs, addressing concerns about high reservation wages among oil shale workers. Only workers who have been employed in the shale oil industry for at least two years are eligible for the wage top-up.³⁴

Larger impacts of the green transition are expected in select occupations across sectors, through changes in the task content of jobs due to cleaner technologies and production processes.³⁵ Without adequate retraining, workers in occupations at high risk of substantial change will not remain as productive and could be at risk of losing their job. For example, construction inspectors, agricultural technicians, and architects are expected to update their skills to apply and be aware of new green production standards and cleaner technologies.

³⁴ The support is about 30 percent of the old salary, but not more than €1,000 per month. The duration of the wage top-up depends on the duration of employment in the oil shale sector. Those who have worked for under 5 years can receive the payments for up to 6 months. Those that have worked for 5–10 years can receive payments for 9 months. Those that have worked for over 10 years are eligible for up to 1 year of payments (https://ec.europa.eu/regional_policy/sources/funding/just-transition-fund/project-fiches/project-fiche-fund-ida-virumaa.pdf).

³⁵ Globally, economic models of transitions to low-carbon economies predict modest net job growth (ILO 2018; IMF 2020; IEA 2022a; Hafstead and Williams 2020). In Europe and Central Asia (including Western Europe), 4.2 percent of total employment is currently in brown jobs and at risk of being lost, while 13 percent of jobs are expected to undergo significant changes in their task content due to the greening of the economy.

5. Social protection to enable just green transitions: practice and evidence

While the previous section focused on social protection programs that can help poor households and workers manage the risks and possible negative impacts of the green transition (the left-hand, blue segment of figure 3.1), this section focuses on social protection programs that can actively contribute to the green transition (the right-hand, green section of figure 3.1). These include programs that (1) equip workers with the skills needed to meet the demand for green jobs and connect workers with employers producing environmentally friendly outputs and promoting environmentally friendly production processes; (2) train and support environmentally sustainable outputs among poor farmers and producers; and (3) promote energy efficient consumption and behaviors among poor people through SPL delivery systems that are integrated with key government agencies implementing subsidies and other green transition policies aimed at reducing pollution and preserving natural resources.

Preparing and connecting workers to green jobs

Meeting the demand for green jobs will require dedicated policies to train workers to compete in this new environment. The appropriate policy to connect workers to green jobs will differ based on the type of green investments and industrial policies and will broadly depend on the green transition plans that central and local governments adopt. For example, investments to expand the renewable energy sectors are associated with increased demand for certain occupations and skills, requiring new courses, updated curricula and scale up within vocational education and training, higher education, and short-term skill programs at all skill levels (i.e., engineers, construction workers, support services solar panel installers). Opportunities exist at all skills level as green skills are wide-ranging, encompassing not only technical and Science, Technology, Engineering, and Mathematics (STEM) expertise but also non-technical, cross-sectoral abilities and those that can be acquired through short training programs (Sabarwal, Venegas Marin, Spivack, & Ambasz, 2024). Green transition policies (regulation and performance standards, for example) that are meant to promote the adoption of green technologies within firms require public support to firms to retrain workers. If policy makers aim to build a workforce that can use green skills to do a wide range of green tasks, an economywide approach that equips people with the right skills is most valuable. This should be accomplished in partnership with the private sector, to identify the exact green skills needed to perform current green tasks and anticipate emerging ones. Public-private collaboration should also extend to job intermediation services to ensure that workers can effectively transition into green jobs and that employers can access the talent they need.

Labor market information and green skills observatories

Understanding which occupations and skills are most in demand to respond to both the short term and medium-term demand for green jobs is at the center of the design of skills development policies, active labor market programs (ALMPs), efforts to support green investments, and public employment services. This requires the development of agreed national concepts and definitions of “green jobs” and the

establishment of a system of data exchange agreements, institutional arrangements and technical solutions to identify, monitor, and forecast the demand for green occupations and skill, and the supply of green skills (i.e., a labor market information system, LMIS [Posadas and Testaverde 2021]). Within LMIS, labor market observatories (LMOs) are institutions with the objective to produce systematic information on occupations and skills demand to help policy makers (and other labor market stakeholders, such as students, employers, job seekers, and training providers) make informed choices. LMOs integrate and consolidate information coming from different sources to produce and disseminate labor market intelligence (Rutkowski 2017; Posadas and Testaverde 2021). In the context of the green transition, LMOs would help answer the following questions: What occupations, education credentials, and skills are needed to make green industries grow? What are the implications of green industrial policy for the structure of sectoral employment?

For example, the India Skill Council for Green Jobs (SCGJ) is an initiative established by the Government of India in 2015 with the objective to guide the implementation of green skills development policies. Its purpose is to define the green industries and to identify the skills needed to implement nationwide, industry-led, collaborative skill development and entrepreneurship initiatives. The SCGJ is responsible for developing green occupation standards and profiles, and providing qualification packages and quality assurance. As of June 2024, 77 green occupational profiles (e.g., solar panel installers, biomass managers, wastewater treatment technicians) and qualification packages to standardize learning outcomes and training assessment have been developed.

The French Ministry of Environment established the French National Observatory for Jobs and Occupations of the Green Economy (Onemev) in 2010 to monitor the employment shifts associated with the expansion of the green economy. Onemev has two approaches to monitor employment: one based on “eco-activities” (those devoted to the production of goods or services for the purpose of protecting the environment or for the sustainable management of resource) and an approach based on tracking the number and type of jobs in nationally defined “green” and “greening” professions.³⁶ Results from Onemev show that between 2004 and 2021, employment in eco-activities increased by 86 percent compared to an average 12 percent total employment growth at the national level. In 2021, activities in the green economy created about 1.2 million full-time equivalent jobs, representing 4.2 percent of the total national employment (Données et études statistiques 2023).

Green skills training programs

While often perceived as highly technical and confined to sectors like energy or transport, green skills encompass a wide range of both technical and nontechnical abilities—many of which can be acquired through short-term, market-responsive training. These types of trainings are part of social protection and labor programs, which can play a vital role in equipping workers—especially vulnerable ones—for greener livelihoods and jobs. At the same time, sustaining and expanding the green skills agenda over the long term requires foundational reforms in the education system, including the development of flexible learning pathways, strong STEM instruction, and better information flows between employers and training institutions (see Sabarwal et al. [2024] for more details). While this paper examines short-term green skill

³⁶ “Green occupations” are those relevant to measuring, preventing, controlling, and mitigating adverse impacts and damage to the environment, whereas “greening occupations” do not have a direct environmental purpose but integrate competences relating to the environmental dimension into their work.

development within social protection systems, it also acknowledges the need for deeper, education-driven reforms to ensure a just and inclusive green transition.

In alignment with the definition of green occupations, India's Ministry of Environment, Forest and Climate Change launched the Green Skill Development Program.³⁷ The program offers green skills courses across various fields through both public and private training providers in 87 locations across the country.³⁸ As of the end of 2024, the program has certified 4,717 trainers and 756 assessors, and has trained 587,303 people since its inception, with the highest number of certifications issued for solar photovoltaic (PV) installers (Suryamitra program) (SCGJ 2024). Based on the feedback received from stakeholders, there is a plan to convert few courses into one-year postgraduate diplomas.

The Regional Network in Energy for Women in the Middle East and North Africa (RENEW MENA) is a coalition launched in 2022 of local, regional, and international partners, including the World Bank, with the objectives to increase women's economic participation in the energy sector value chain. Specifically, it aims to increase women participation in jobs related to the clean energy transition, encourage better workplace conditions for women in both the private and public sectors, and combat widespread gender stereotypes about women's role in STEM fields and increase their visibility in the energy sector in Egypt, Jordan, Iraq, Lebanon, Morocco, West Bank and Gaza, Tunisia, and Yemen (World Bank 2022a). By the end of 2024, RENEW MENA partners have implemented STEM outreach sessions, study tours, internships, job placement services, mentorship, and professional development workshops for about 4,500 women in different countries (RENEW MENA 2024).

The Ghana Youth Employment Agency provides various training modules designed to equip youth with skills in different sectors, including training modules on waste and sanitation, and agriculture and afforestation. The agriculture and afforestation module was launched in 2018 with the core function of planting and tending to trees (Ghana Youth Employment Agency n.d.). Beneficiaries are selected through a rigorous online process to then be matched with training modules, enrolled for two years and provided with a monthly allowance during the training. About 50,000 youth were enrolled in the agriculture and afforestation module, though 35,000 people completed the training in 2020 due to resignations (Forestry Commission 2021).

Green skills training initiatives are often developed at the subnational level to meet local labor market demand. The development of green jobs is a priority in the Santa Fe Decent Work Agenda 2017–20. Argentina's province of Santa Fe launched the provincial program "Empleos Verdes—Cuidar el planeta da trabajo" in 2016 to train job seekers in green jobs such solar panel installers, bioenergy workers, and integral waste managers. The short-term training courses were carried out in collaboration with municipalities, communities, and civil society organizations (Saget, Vogt-Schilb, and Luu 2020). As of 2018, 52 courses had been held with an approximate duration varying between 80 and 120 hours for about 1,500

³⁷<https://moef.gov.in/green-skill-development-program#:~:text=Skill%20development%20is%20one%20of,of%20reaping%20the%20demographic%20dividend.>

³⁸ These include: pollution monitoring (air/water/soil); emission inventory; sewage treatment plant (STP)/effluent treatment plant (ETP)/common effluent treatment plant (CETP) operation and maintenance; waste management; Environmental Impact Assessments; forest fire management; water budgeting and auditing; conservation of river dolphins; wildlife management; para taxonomy including Peoples Bio-Diversity Register (PBR); mangrove conservation; bamboo management and livelihood generation; valuation of ecosystem services; and water budgeting and auditing, among others.

trainees (SantaFe Provincia 2018). In 2023, the central government of Argentina launched the Empleo Verde program to boost the growth of green employment and encourage economic activities managing the use of natural resources.³⁹ The program supports vocational training for occupational profiles that are environmentally sustainable.

Intermediation and job matching services for employers in green industries

The Public Employment Services in Slovenia, in cooperation with the Ministry of Environment and Space, introduced a point system to define green jobs based on employers' description of the occupation, output produced, and compliance with green certificates and standards. The Partnership with the Ministry of Environment ensured alignment of the point system with broader climate and environmental goals and effective implementation. Vacancies and employers are rated according to the point system to facilitate job seekers and the unemployed access to green employers. The point system is also used to target hiring incentives (€680 per unemployed hired per month for a defined period) to green employers. This new employment incentive called "Green Jobs" was launched in 2021 (European Commission 2023a). The results so far show that 323 companies and 386 unemployed were matched through the program. About a quarter of participants—mostly male—were first-time labor market entrants, 15 percent were under 25 years of age, 17 percent were long-term unemployed, 5 percent had a disability, and 10 percent were social assistance beneficiaries (Tomic et al. 2025). Green occupations most commonly subsidized include technicians for the efficient use of natural resources, production or installation of technologies to produce energy from renewable sources, increasing energy efficiency, regulating or rehabilitating the environment, waste management and reuse, green tourism, and sustainable mobility.

Wage subsidies for green jobs

To stimulate the provision of urban recycling services, the Brazilian state government of Minas Gerais introduced the Recycling Bonus program (Bolsa Reciclagem) in 2011. The bonus consists of a financial payment to associations and cooperatives based on the quality and type of recyclables collected.⁴⁰ The bonus benefitted 2,000 waste pickers organized in 137 associations and cooperatives in 2018 and contributed to the recovery of approximately 40,000 tons of waste. This benefit has been complemented by the "recycling with dignity" state benefit (Reciclando Dignidade) in 2024 to ensure that recyclable material collectors—members of the cooperatives—are covered by old age pension and social insurance for temporary disability benefits, maternity leave, survivor's pensions, etc. The "recycling with dignity" benefit is expected to subsidize social security contributions for at least 1,474 recyclable material collectors for four years starting January 2025 and covering 64 municipalities (Dias 2016; Meioambiente 2024).

³⁹ <https://www.argentina.gob.ar/trabajo/empleo-verde>; <https://www.argentina.gob.ar/programa-empleo-verde>.

⁴⁰ The associations and cooperatives that collect and sell a higher quantity of recyclable materials (paper, plastics, metal, glass) receive a greater transfer. As of 2020, the material collected in greatest quantity was paper. Eligibility conditions for cooperatives to receive the bonus include that they be legally established for more than a year, employ no minors under the age of 18 years, and that all children of members be enrolled in school. Dashboard of Bolsa Reciclagem: <https://app.powerbi.com/view?r=eyJrIjoimM2VhYzY2NjAtMTg5My00NWE1LThtNmMtMjI1NzQ0NDZhM2QwliwidCI6IjEyN2Y2ZDU1LTA1NjgtNDhkZS05YzJhLWE5ZmQxZTMwYjk0MSJ9>.

Incentives for green business start-ups

Entrepreneurship support programs could contribute to green job creation when well targeted to economic activities addressing environmental issues. In middle- and high-income countries with established public employment services offering a wide menu of ALMPs, entrepreneurship support programs are targeting financial and technical support for business start-up to economic activities identified as a priority to facilitate the green transition. For example, the Croatian Public Employment Service introduced a financial grant for registered unemployed willing to start a new green business.⁴¹ Green businesses are defined as businesses (1) producing environmentally friendly goods and services, contributing to the conservation of natural resources; (2) providing education and skill training related to green technologies and practices; (3) assessing compliance with environmental standards and the implementation of environmental regulations and standards; and (4) increasing the general level of public awareness of environmental issues. The implementation of this new measure has highlighted the need to further define eligible green businesses (European Commission 2023a).

In lower-income country contexts, micro-entrepreneurship support can promote green job creation among poor people when the in-kind assets, financial capital, and training interventions are designed to promote environmental conservation and restoration. The Kenya National Youth Opportunities Towards Advancement (NYOTA) project is a comprehensive youth employment program implemented by the Ministry of Youth Affairs, Creative Economy and Sports; the Ministry of Labor and Social Protection; and the Ministry of Cooperatives, Micro, Small and Medium Enterprises in partnership with the World Bank. While at its inception stage, the NYOTA program is designed to provide soft skills, on-the-job training, job matching services, business start-up grants, and business development services to about 800,000 vulnerable youth by 2029. Promoting youth employment in the green or blue economy is part of the program's objective: digital and other skills demanded in green and blue economy will also be delivered to help youth support green investments either as employees or as entrepreneurs.

Other indirect measures to foster green innovation and entrepreneurship include business support services, incubators, and mentor networks for new green entrepreneurs. The Ghana Climate Innovation Center was launched in 2016 with the support of the World Bank and other donors to provide support to small and medium enterprises and start-ups in developing business concepts for climate-smart agriculture, domestic waste management, energy efficiency, water management, and solar panels (World Bank 2016; Ghana Climate Innovation Centre n.d.). The center is managed by a consortium comprising the Ashesi University College, Ernst & Young, SNV Ghana, and the United Nations University Institute for Natural Resources in Africa. For example, the Business Incubation Program empowers entrepreneurs with innovative business solutions by providing six to nine months' trainings.

Global skills partnerships for green jobs

The green transition may be constrained by the lack of workers and skills needed to grow specific green industries. Labor and skills shortages in some high-income countries are affecting the implementation of green technologies despite the potential for job creation through the green transition. In such country contexts, migration policies and bilateral and global skills partnerships can help address labor shortages. Global skill partnerships offer a model in which origin and destination countries jointly finance skills

⁴¹ Start-up grants for unemployed registered with PES are up to €15,912 to cover start-up costs. Beneficiaries are obliged to work for at least 24 months full-time in their business and submit evidence of the eligible costs.

training in the origin country for skills in-demand in both countries, while providing legal migration pathways to a subset of graduates. They have been successfully tested in various sectors and countries and can be tailored to train workers in origin countries to support the green transition both in origin and receiving countries (Huckstep and Dempster 2024; Acosta et al. 2025).

For example, high workforce shortages are hindering Germany's solar PV industry: the country will need 165,000 solar workers by 2026 to meet the annual installation target, and 60 percent of electrical contractors working in solar had vacancies as of February 2023 (Huckstep and Dempster 2024). To address these labor shortages, Germany has sought to attract foreign workers by reforming migration policies and through bilateral partnerships for training and active recruitment, in addition to domestic skills training initiatives. In February 2023 the German Solar Association (BSW) has agreed a field-leading partnership with India's Skill Council for Green Jobs, allowing around 2,000 Indian solar PV installers to work in Germany's solar PV sector from 2024 to 2026. This agreement addresses India's surplus of workers and unemployed graduates from SCGJ programs due to a lower-than-expected PV installation rate in India due to high import costs for solar PV panels. This will allow Indian workers to develop and enhance their skills, and some may return to address skill requirements in their home country when the demand for labor in solar PV panel production increases. The agreement between BSW and SCGJ is a business-to-business agreement but is being managed in collaboration with government departments and associations through the Hand in Hand for International Talents project (Huckstep and Dempster 2024; Yuen 2023).

Similarly, Australia is experiencing labor shortages as it seeks to increase its share of low-carbon power generation from 32 percent in 2023 to 82 percent in 2030. Specifically, 168,000 workers are needed by 2050, in addition to those expected to transition from fossil fuel sectors (Huckstep and Dempster 2024). While migration alone will not address the labor shortages to transition to a clean energy economy, a well-designed and fit-for-purpose skilled migration program is part of the solution (Jobs and Skills Australia 2023). Existing initiatives—such as the Australia Pacific Training Coalition (APTC)—could be expanded to increase the supply of workers with vocational and technical skills needed in the low-carbon energy sector.⁴² To date, hundreds of Fijians have been trained in field of study relevant to the green transition (such as heating, ventilation, and air conditioning and electronics) and in compliance with Australian qualification standards (Huckstep and Dempster 2024). New initiatives and partnerships are also emerging. In November 2024, Australia and India established the India-Australia Renewable Energy Partnership, which aims to build the future workforce for the renewable energy sector through skills and training.⁴³

Promoting environmentally sustainable livelihoods

Environmental conditional cash transfers

Environmental conditional cash transfers (ECCTs) are designed to simultaneously fulfill social protection and environmental objectives. Recipients receive cash transfers conditional on complying with sustainable environmental and agro-ecological practices (e.g., climate-smart soil management or sustainable agroforestry), similar to Payment for Ecosystem Services (box 5.1). The targeting of ECCT programs usually

⁴² It has trained 20,000 full qualification graduates in different sectors (not only green) since inception, and 8 percent have migrated to Australia since 2019 (Acosta et al. 2025).

⁴³ <https://www.dccew.gov.au/sites/default/files/documents/establishment-india-australia-renewable-energy-partnership.pdf>.

prioritizes households or communities based on socioeconomic criteria, such as income levels or vulnerability, in addition to environmental considerations.

Paraguay's Poverty, Reforestation, Energy and Climate Change (PROEZA) project offers an example of ECCTs. PROEZA started in 2019 with the objective of encouraging households to adopt low-emission and environmentally sustainable production methods in food production. The program is coordinated by an inter-institutional executive committee including the six line agencies responsible for agriculture, forestry, social protection, indigenous peoples, environment, and energy (Green Climate Fund 2018). The project targets poor and extremely poor household beneficiaries of the government's conditional cash transfer program *Sembrando Oportunidades*⁴⁴ (Györi, Diekmann, and Kühne 2021) and provides top-up benefits and expert technical support to eligible households⁴⁵ to incentivize the establishment of agroforestry activities in their land, close-to-nature forest plantations, and natural forest regeneration activities (FAO 2023). The top-up benefits are paid to households who have managed a productive agroforestry model conditional on meeting agreed key performance indicators such as the survival rate of plants, the species mix and composition, and pruning activities, among others. The program also provides concessional credit to landowners (medium-sized holdings) to encourage investments in reforestation (FAO 2023). As of 2023, 3,418 people have directly benefited since the start of the project and many have incorporated self-consumption crops in their agroforestry plantations (Green Climate Fund 2024).

The Brazilian *Bolsa Verde* program,⁴⁶ which provides top-up cash transfers to beneficiaries of the *Bolsa Família* program conditional on them performing environmental conservation and restoration activities, offers another example of ECCTs.⁴⁷ The program started in 2011 and targeted the Amazon region, but later expanded to the whole country. Eligible families must reside in specific environmentally sensitive areas recognized for their ecological importance, be registered as recipients of *Bolsa Família*, and carry out activities of conservation and sustainable land use. *Bolsa Verde* provided cash transfers to around 74,522

⁴⁴ *Sembrando Oportunidades* consolidates several programs, including Tekoporã, which is a conditional cash transfer program for families with school-aged children; Tenonderã, which provides training and asset transfers; and Adultos Mayores, which offers pensions for the elderly.

⁴⁵ Eligibility criteria include: (1) participation in the Tekoporã CCT program; (2) being geographically located in the agreed areas with high degree of social vulnerability; (3) being located in an area with high degree of climate vulnerability, deforestation risk, land degradation, low vegetation coverage, and close to an industry using firewood fuel; (4) having a minimum of 0.8 hectares of formal land ownership to implement agroforestry conditions; and (5) having a track record of land ownership maintained in a peaceful manner (without disputes) and land formal registration according to the Agrarian Law of Paraguay. Female-headed or indigenous households have preferential access to the E-program.

⁴⁶ *Bolsa Verde* Program, officially known as the Programa de Apoio à Conservação Ambiental (Program for Environmental Conservation Support), is coordinated, implemented and operated by the Ministry of Environment (MMA). However, is subject to deliberations by the program's Steering Committee, formed by the Presidency's Chief of Staff Office (CC-PR), the Special Secretariat for Family Agriculture and Agrarian Development at the Presidency's Chief of Staff Office (SEAD-PR), the Ministry of Social Development (MDS), the Ministry of Finance (MF), and the Ministry of Planning, Development and Management (MPDG) (WWP 2017).

⁴⁷ While the program initially targeted only the Amazon, it was then expanded to the entire country. Targeting was based on: (1) geography—families living in areas and carrying out activities of conservation and sustainable use, as identified by the Chico Mendes Institute for the Conservation of Biodiversity (ICMBio), the National Institute of Colonization and Agrarian Reform (INCRA), and the Secretary of the Patrimony of the Union (SPU); and a (2) means test—households living in extreme poverty (monthly per capita income below R\$85), registered in CadÚnico as recipients of the *Bolsa Família* Program (<https://dds.cepal.org/bpsnc/programa?id=60>).

extremely low-income Brazilian households to maintain forest cover (OXFAM 2020). Recent quantitative research suggests that the *Bolsa Verde* reduced annual deforestation by 0.18 percentage point of the area size (Wong, Kuralbayeva, and Anderson 2024). This suggests that the measure was cost-effective in terms of carbon reduction.⁴⁸ Further, qualitative evidence suggests that the program contributed to increased income and livelihoods (OXFAM 2020). The program was deactivated in 2016 and was then reinstated in 2023 with increased quarterly payments to households (Ministry of Environment and Climate Change n.d.).

Recent studies show that some conditional cash transfers targeted at poverty alleviation can reduce deforestation even when they have no explicit condition on the environment. For example, the *Keluarga Harapan* (“Family Hopes”) cash transfer program, abbreviated as PKH, in Indonesia reduced tree cover loss in villages by 30 percent without having any defined conditions with conservation activities (Ferraro and Simorangkir 2020).⁴⁹ In Colombia, households receiving cash transfer increase the purchase of land and energy-intensive goods but municipalities receiving conditional cash transfers show lower deforestation with respect to municipalities that did not benefit from the program (Malerba 2020).

Box 5.1 Payments for environmental services

Payments for environmental services (PES), sometimes also referred to as payments for ecosystem services, aim to incentivize landowners and resource users to manage land for specific environmental benefits, such as carbon sequestration, watershed protection, and biodiversity conservation. PES involve voluntary, conditional agreements between buyers and sellers over a defined environmental service. These payments are often outcome based and the programs target areas where interventions can yield significant ecological benefits, contributing to poverty alleviation and livelihood diversification.

PES and environmental conditional cash transfers (ECCTs) are similar as they both aim at improving environmental and socioeconomic outcomes; however, they differ in their primary objective and targeting. While PES have as a primary objective the improvement of ecosystems, ECCTs typically focus on social objectives, such as poverty alleviation or improving human capital, and their targeting may prioritize households or communities based on socioeconomic criteria, such as income levels or vulnerability, in addition to environmental considerations.

Costa Rica’s *Pago por Servicio Ambiental* is a pioneering PES program providing direct payments to landowners for sustainable land-use and forest management. Managed by the National Forestry Financing Fund (Fondo Nacional de Financiamiento Forestal, FONAFIFO), it is funded through fuel taxes, water charges, biodiversity conservation certificates, carbon credits, and public-private partnerships. Landowners receive direct payments in exchange for the environmental services generated by their land through the adoption of sustainable land-use and forest management practices. From 1997 to 2019, over 18,000 families benefited, including indigenous communities and family associations. The program has evolved to include women and small farmers. Indigenous territories have special conditions for program entry, and about 100,000 indigenous people in total benefit from the incentives

⁴⁸ Valuing the carbon sequestration of the saved forest at the estimated social cost of carbon (SCC) for Brazil at US\$24 per tonne of CO₂, research finds that the program removed carbon worth US\$161 million, about 1.75 times the calculated program costs (Wong et al. 2025).

⁴⁹ PKH transfers cash conditional on households meeting health and education obligations. Transfers are approximately 15 to 20 percent of recipients’ consumption and are received quarterly for six to nine years.

received through the program (UNFCCC 2023). Funds are used for education, health care, food, and community projects.

The Mexico Payments for Hydrological Environmental Services program, launched in 2003, compensates landowners for conserving forests crucial for water supply and quality. Initially focused on areas at risk of overexploitation and deforestation, it later expanded to include ecosystem services like carbon storage and biodiversity. In 2006, it became part of a broader program—Pago Por Servicios Ambientales—with an increased focus on poverty reduction. In addition to the geographic targeting, socioeconomic criteria related to income and asset ownership, gender, and being part of the indigenous population were added to the targeting rules. About 78 percent of payments went to marginalized communities (Porras and Asquith 2018). The program involves contracts with the National Forestry Commission (Comisión Nacional Forestal, CONAFOR), offering fixed payments for sustainable practices, contingent on preventing deforestation. Studies show it reduces land cover loss by 40–51 percent and modestly alleviates poverty, with greater environmental benefits in low-poverty areas and household gains where deforestation risk is low, illustrating the challenge of meeting multiple policy goals (Alix-Garcia, Sims, and Yañez-Pagans 2015).

In Brazil, Bolsa Floresta provided financial support to households living within state-run conservation units in Amazonas conditional on meeting certain environmental conditions.⁵⁰ One component consisted of an annual productive investment to finance community-based sustainable development reserves in line with the protected area's management plan. The family component consisted of a monthly cash transfer to eligible households (paid to the female spouse), conditional on a signed agreement that commits the household to good forest management practices, including net-zero deforestation. The program was launched in 2007 and run by the nongovernmental organization Amazonas Sustainability Foundation and restructured in 2022 into a new program called “Forest Guardians” which, among others, increases compensation to families (Bakkegaard and Wunder 2014.; GCF Task Force 2022).

Green public works

Green public works combine environmental objectives with the consumption smoothing or employment objective through cash-for-work and temporary employment. The environmental objective is reflected in the selection of the subprojects that would help address environmental degradation. Examples of green public works subprojects include preserving wetlands, restoring forests, protecting coral reefs, rehabilitating riverbanks and coastal banks, and creating public infrastructure that protects against both chronic climate stressors and acute climate impacts such as flood barriers and stormwater management systems. Balancing local needs with environmental and social objectives can be achieved through locally led participatory planning processes, through which communities identify and prioritize subprojects that respond to local needs (World Bank, forthcoming).

When they are large and provide regular support to households within the same geographic areas over time, green public works have the potential to meet climatic and environmental goals on a large scale. For example, India's Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS) and

⁵⁰More than 9,000 families had benefitted from Bolsa Floresta, and the target is higher for the new Forest Guardians program (14,500 families) (World Bank 2024a).

Ethiopia's Productive Safety Net Program (PSNP) have contributed to reduce environmental degradation on a large scale. Activities within the MGNREGS have contributed to better groundwater recharge, increased soil fertility, and reclamation of degraded lands (Behrer and Pullabhotla 2024; Deininger and Liu 2019). Similarly, the PSNP focuses its subprojects on soil and water conservation in the highlands and rangeland management activities in the lowlands (about 60 percent of the subprojects) to address land productivity losses due to erosion and unsustainable land use practices. There is evidence that Ethiopia's PSNP has contributed to increased tree cover and thus increase in carbon sequestration (Hiroven et al. 2022).

Malawi's climate-smart enhanced public works⁵¹ and Fiji's Jobs for Nature 2.0 are other examples of green public works. Malawi's climate-smart enhanced public works program has been scaled up since 2022 based on positive qualitative evidence on trees, as well as soil and water conservation during the pilot phase (Györi, Diekmann, and Kühne 2021). Public works⁵² beneficiaries are trained on the technical aspects of soil and water conservation, riverbank protection, and forestry establishment and management, and how to execute the works. This program covers approximately 13 percent of the population⁵³ and reflects Malawi's efforts toward climate adaptation and mitigation (World Bank, forthcoming). The Jobs for Nature 2.0 program in Fiji aims to create rural employment opportunities while mitigating environmental degradation. Participants engage in various activities including mangrove and seagrass planting, forest restoration to support wetlands protection, riverbank rehabilitation, and biodiversity enhancement (World Bank, forthcoming).

While green public works offer substantial benefits, there are also potential challenges, given the need to balance environmental objectives with those for poverty reduction or employment promotion. Integrating climate and environmental goals into traditional public works programs can be complex due to the need for specialized environmental data and technical expertise, which may not always be readily available. Further, improper implementation can lead to unintended negative consequences. Successful green public works also require extensive collaboration among various institutional and policy stakeholders. Finally, the impact of green public works on environmental outcomes is rarely measured, and their cost-effectiveness compared to traditional public works programs remains uncertain. A more detailed discussion on the challenges to balance social and environmental objectives is in World Bank (forthcoming).

Economic inclusion programs to preserve and restore natural resources

Some economic inclusion (EI) programs support income-generating activities among poor people with the objective to preserve and restore natural resources, particularly in areas where ecosystems are under pressure from overuse (e.g., nontimber forest products) (Costella et al. 2023; Arévalo-Sánchez et al. 2024). In many contexts these EI programs are part of environmental projects aimed at restoring or protecting

⁵¹ Supported through the World Bank Malawi Social Support for Resilient livelihood project (P169198).

⁵² Subprojects for forest restoration, through both tree planting and natural regeneration, land resources conservation initiatives including run-off management activities such as construction of infiltration pits, marker ridges, swales, planting of vetiver grass and gully reclamation among others, and rehabilitation of water supply, drainage, and small access roads

⁵³ As of January 2025, 1.9 million poor people have been supported with climate-smart livelihoods, 25,000 SLG formed, over 1.9 million poor people trained in basic livelihoods, and 10,000 groups supported with productive grants and linked to microfinance and markets. In addition, 2004 youths were supported with value addition skills support and 1,600 groups benefited from Legumes Enterprise Structured Program.

particular ecosystems. For example, in Chad the Albia-Chad Local Development project supported by the World Bank supports agricultural activities that contribute to land restoration through training in management of water and natural resources, advisory services, targeting in particular extremely poor, women-headed households (based on a rapid poverty level diagnostic).⁵⁴

The Malawi Social Support for Resilient Livelihoods Project (SSRLP) provides training and in-kind transfers to support sustainable livelihoods for poor households, by integrating climate-smart agricultural practices. As of January 2025, over 1.9 million poor people were trained in basic livelihoods, and 10,000 groups supported with productive grants and linked to microfinance and markets. The participants are trained on the technical aspects of soil and water conservation, riverbank protection, and forestry establishment and management, and how to execute the works.

A new component to the Ethiopia PSNP provides training and business grants to help public works beneficiaries start climate-sensitive income-generating activities (on and off farm) to diversify livelihoods such as watershed management, small-scale irrigation, and beekeeping in protected areas. As of December 2024, 8,560 beneficiaries have been supported with a livelihood grant.

Leveraging SPL delivery systems to enhance other sectoral green transition policies

Social protection delivery systems have the potential to enhance the impact of green transition policies by stimulating the demand for energy efficient investments and by increasing the awareness and take-up of incentives and subsidies to produce and consume environmentally friendly goods and services, especially among poor, marginalized and vulnerable subpopulation groups. SPL delivery systems—registries, payment platforms, ID systems—offer governments ready-made tools to identify, reach, and serve different types of beneficiaries, helping, for example, to target energy subsidies and messaging on changing energy inefficient and environmentally harmful consumption behaviors.

Flagship cash transfer and other social protection programs could serve as a platform to provide information on environmentally friendly consumption behaviors and to inform choices to live and travel in more energy-efficient ways, therefore enhancing the impact on carbon footprints. To do so, social registries, program specific beneficiary registries, payment systems, and identification systems need to be interoperable with the administration systems of government agencies implementing green policies and programs, such that these data can be used to channel such information, which would need to be tailored to the target audience.

Examples are scarce, indicating the potential for further exploration and scale-up. Given below are some examples of energy, transport, agriculture, and forestry programs promoting the green transition by

⁵⁴ This supports the setting up of an integrated agricultural production site—a fenced area including agroforestry, market gardening, storage warehouse, training centres (provision of advisory services), and water points—in each selected hotspot outside the reserve. In these sites, producers develop activities to improve agricultural production and marketing through the adoption of climate-smart agriculture technologies. Individual innovators (champions) are identified to help manage the demonstration plots inside each site where all beneficiaries will learn how to apply theoretical knowledge learned during training sessions

leveraging (and with the potential to leverage) the social protection delivery system to maximize their policy impacts.

Air pollution management through residential heating and cooking

Residential heating is the single-largest source of air emissions in Poland, home to 36 of the 50 most polluted cities in Europe. The government took several steps to reduce air pollution in the form of grants and tax relief for energy efficiency retrofits and replacement of heating sources through the Clean Air Priority Program (CAPP).⁵⁵ Take-up of the CAPP is low among poorer households, who also tend to live in uninsulated buildings with outdated heating, requiring greater investments. Due to the poor state of housing, these households consume more energy, and therefore spend larger shares of their budget on utilities (energy poor) and have limited capacity to undertake energy efficient investments, even if subsidized. These households are more likely to use free fuel—trash as well as wood from forests—for heating and make other types of investments. Amendments are in progress under the CAPP that aim to help vulnerable households by supporting the preparation and implementation of the investments and strengthening the precision of targeting in the lowest-income group. However, the success of those actions will be heavily dependent on the engagement of institutions and municipalities (World Bank 2024d).

In the United Kingdom, health care providers and social workers are part of the Cold-Health Alerts prevention and response that connect vulnerable people with existing programs to increase energy efficiency such as the Boiler Upgrade Scheme and the Warm Homes.⁵⁶ Similarly, in Scotland, social workers providing long-term care services carry-out energy inspections of old people's homes and help them access grants for energy upgrades (Cook 2020).

In Mongolia, a stove subsidy program has been implemented in the past decade to promote the use of energy-efficient stoves, aiming to reduce air pollution and improve health outcomes. The program, funded by international organizations such as the World Bank, the Millennium Challenge Corporation, and bilateral donor support, provides subsidies and awareness campaigns to encourage households to switch from traditional coal-burning stoves to more efficient alternatives. Initially it started as the Clean Stove Initiative (CSI), launched in 2012 with the support of the World Bank to combat severe air pollution in Ulaanbaatar, primarily caused by coal-burning stoves in traditional *ger* districts (Zhang and Adams 2016).⁵⁷ The World Bank project provided clean stoves to 40,813 households between 2013 and 2015⁵⁸ and contributed to lower particulate matter emissions in Ulaanbaatar city. Electric thermal storage (ETS)

⁵⁵ Launched in 2018, Poland's Clean Air Priority Program (CAPP) supports thermo-modernization through energy efficiency retrofits and replacement of heating sources such as electric heating systems and gas boilers as well as air source and ground source heat pumps.

⁵⁶ <https://www.sgn.co.uk/about-us/vcma/alliance-scotland>

⁵⁷ This is part of the broader EAP Clean Stove Initiative (CSI), which focuses on achieving access to modern cooking and heating solutions in the EAP region, particularly scaled-up access to advanced cooking and heating stoves for poor, primarily rural households who will use solid fuels to meet most of their cooking and heating needs beyond 2030. The EAP CSI is a multi-country, multiphase program with funding support provided by Australian Aid (formerly AusAID) and the World Bank's Asia Sustainable and Alternative Energy Program (ASTAE) and Energy (Zhang and Adams 2016).

⁵⁸ The project also distributed Electric Thermal Storage (ETS) heaters to small and low-income households under a rent-to-own model, where households paid a fixed monthly fee for a set number of months before gaining ownership. However, some households were unable to afford the cost of ETS heaters and the additional insulation cost, even with the rent-to-own scheme and a 70 percent subsidy for insulation, and the final number of households achieved was 280.

heaters were also distributed, under a rent-to-own model. However, many households proved unable to afford the costs of the ETS heaters, and the heaters were only allocated to 280 households (World Bank 2024c). While the project did not use social protection delivery systems, social registry could be used to reach out to vulnerable populations and possibly incentivize ETS investments.

The Pradhan Mantri Ujjwala Yojana (PMUY), launched in 2016 by India's Ministry of Petroleum, aims to accelerate the switch from biomass/kerosene to clean liquefied petroleum gas (LPG) cooking fuel by providing poor women free LPG connections and subsidized first refills (Yadav 2020). The program used Socio Economic and Caste Census registry data (SECC 2011) to identify eligible beneficiaries. Poor women from households meeting the poverty criteria in the social registry were selected and received a subsidy covering the connection cost. The subsidy was delivered by the public oil companies to their bank accounts leveraging the direct benefit transfer payment system.⁵⁹ The poverty targeting proved highly effective: by 2019, PMUY had covered the cost of over 80 million LPG connections to poor women across India (Nandwani and Jain 2024). However, audits have revealed inconsistencies between SECC data and actual beneficiary information, highlighting the need for regular data updates and verification (Ministry of Petroleum and Natural Gas 2019).

The US Low Income Home Energy Assistance Program (LIHEAP) and the Weatherization Assistance Program (WAP) are federally funded energy program designed to assist poor households with repairs and replacement of heating and cooling systems. The funding is distributed to states that are responsible for managing the program. Eligibility is generally based on household size, residency, income, heating costs, and any relevant medical conditions. For example, in the State of New York, LIHEAP eligibility criteria include the participation in other social assistance programs such as the Supplemental Nutrition Assistance Program, the Temporary Assistance program, or the Supplemental Security Income (SSI). The program managed by the local social service department includes different components⁶⁰ to increase energy efficiency and covers the costs of replacement and repairs of heating and cooling equipment, and utility expenses. The benefits are paid directly to the heating vendor providing repairs and replacements, or to the utility company. Similarly, the WAP⁶¹ provides services to low-income households such as insulation, duct sealing, heating, and cooling systems repairs or replacement, air infiltration mitigation, roof repair, mold remediation, and installing energy efficient lighting and appliances. In the District of Columbia, eligible households must have annual income 60 percent below the state median Income, or at least one household member should be receiving Temporary Assistance for Needy Families (TANF) or SSI benefits.⁶²

⁵⁹ The Direct Benefit Transfer is a system to transfer subsidies directly to people through their linked bank accounts implemented by the Government of India since 2013 (Press Information Bureau 2022).

⁶⁰ The components include: the Emergency HEAP benefits to address a heating crisis during the winter months; the HEAP Clean and Tune benefit to assist in the cleaning of primary heating equipment or chimney cleaning, minor repairs, or installation of carbon monoxide detectors, to allow safe, proper, and efficient operation of the heating equipment; the Heating Equipment Repair or Replacement component of to help eligible low-income homeowners repair or replace furnaces, boilers and other direct heating components; the HEAP cooling assistance program targeted to households with medical conditions providing assistance to low income households with the purchase and installation of an air conditioner or a fan.

⁶¹ The program offers weatherization services to approximately 32,000 homes annually with DOE funding. Since its inception in 1976, WAP has provided weatherization services to over 7.2 million families.

⁶² <https://doee.dc.gov/service/wap>.

Promoting green agriculture and forest conservation

China's Green Agricultural and Rural Revitalization project supported by the World Bank aims to enhance the adoption of environmentally sustainable agriculture in selected areas of Hubei and Hunan. Key results areas include the greening of selected agricultural value chains and expanding access to rural solid waste and wastewater services. The lack of specific technical skills among farmers, extension workers, and rural cooperatives is recognized as among the key obstacles to the adoption of green technologies and waste management practices. To address the skills gap, the project supports the provision of technical training and capacity building to fully master the details of good agricultural practices as well as training on operations and maintenance of wastewater and solid waste management systems. While the development and delivery of training modules will be outsourced under the project, there is scope to leverage the training provision of the Ministry of Human Resources and Social Security (MOHRSS) during implementation. MOHRSS organizes regular short-term training for farmers and returning migrant workers that have the potential to be enhanced and coordinated with the five-year program recently launched by the Ministry of Agriculture and Rural Affairs to develop a new type of professional farmer equipped with a wide range of skills, from soil management and crop and livestock/poultry production to marketing and business planning (World Bank 2023a).

As in other African countries, cash transfer programs are primarily delivered through mobile money in Côte d'Ivoire. Social protection mobile payment systems have the potential to be leveraged to redistribute benefits from emission reductions from forest conservation, forest rehabilitation, and agroforestry activities, particularly so given that the majority of the population owns a mobile phone. For instance, Côte d'Ivoire received US\$35 million from the World Bank Forest Carbon Partnership Facility to reduce carbon emissions. Payments are based on verified emission reductions from forest restoration, conservation, and reduced deforestation through sustainable agriculture, part of the Reducing Emissions from Deforestation and Forest Degradation (REDD+) mechanism.⁶³ Direct beneficiaries include forest-dependent communities such as cooperatives or organizations of cocoa producers. Payments to members of these communities are made through direct transfers to their mobile money accounts by a mobile money company, unless the beneficiary requests another mode of payment (Ministry of the Environment and Sustainable Development 2023).⁶⁴

⁶³ Under the framework with these REDD+ activities, developing countries can receive results-based payments for emission reductions when they reduce deforestation (<https://unfccc.int/topics/land-use/workstreams/redd/what-is-redd>).

⁶⁴ The beneficiary must have an identification card and bank account number or a mobile payment service number (Mobile Money) registered in their name. REDD+ Benefit Sharing Plan: https://www.forestcarbonpartnership.org/sites/default/files/documents/cote_d_ivoire_final_benefit_sharing_plan_updated_december_12_2023.pdf.

6. Conclusions

Social protection and labor (SPL) policies are being harnessed globally to support a just green transition, enabling workers to seize new opportunities, while also protecting workers and people from potential job and income losses. Although international environmental agreements began in the late 20th century,, significant milestones, as set out in the Paris Agreement, such as the Nationally Determined Contributions (NDCs), were only established in 2015. Consequently, many green transition policies are recent, and empirical studies on their distributional impacts are still emerging.

The literature on relying on SPL policies to mitigate these impacts and prepare workers for the green transition is limited, especially regarding empirical evaluations of their effectiveness (for instance on the optimal size and duration of complementary SPL policies). This paper reviews existing country cases to provide policy and operational insights for integrating SPL into NDCs, promoting environmentally sustainable and inclusive growth strategies and identifying options for adapting existing SPL programs or introducing new ones to promote environmental objectives.

The conceptual framework in this paper demonstrates how social protection and employment programs can shield people and workers from the adverse effects of green transition policies or help them capitalize on new opportunities. Specifically, SPL responses can support vulnerable households in coping with price increases due to green transition policies, such as through cash transfers complementing fuel subsidy reforms; manage disruptions to natural resource dependent livelihoods in household enterprises and smallholder producers, such as through in-kind and cash transfers and extending public work programs during temporary fishing closures; and address job losses and changing job tasks due to firms' adoption of cleaner technology and practices, for example, through upskilling and reskilling during coal mine closures. Additionally, public employment services, skills training and employment programs can help workers take advantage of opportunities arising from the green transition by preparing and connecting them to green jobs. Economic inclusion programs, green public works, ECCT and incentives for green business start-up could play a key role to promote environmentally sustainable livelihoods, especially when designed in coordination with other line ministries, for example ministries of agriculture and environment. A less explored and emerging area is the potential to leverage SPL delivery systems (i.e., identification, social registry, targeting, and payment systems) to raise awareness and enhance the effectiveness of green transition policies in energy, transport, and agriculture for example.

Case studies reveal that all social protection instruments—across social assistance, social insurance, and labor market policies—can be employed to achieve these goals, with their selection and potential modification depending on the context. While all three types of instruments can be contextually applied, social assistance and labor market policies are currently more frequently used to complement green transition policies than social insurance. Social assistance, such as cash or in-kind transfers, can respond to short-term livelihood shocks, as seen in fishery closures, or to address price increases affecting poor households, as in energy subsidy reforms. Severance payments and unemployment insurance provide income support in cases of job losses due to reduced demand from polluting industries and shifts in job tasks, exemplified by coal mine closures. For affected workers who are more difficult to reemploy or who are close to retirement age, unemployment insurance

could be complemented by temporary unemployment assistance and extended paid leave. For uninsured workers, temporary social assistance and voluntary saving accounts offer income protection. Active labor market policies can help workers cope with negative green transition policy effects, such as coal mine closures, and seize opportunities from green transition, like increased demand for green jobs and skills in renewable energies, although their design may need to be modified with this objective in mind.

To complement green transition policies, social protection and labor programs can be implemented through either new measures or by extending and modifying existing programs. For instance, recent upskilling and incentive programs, such as Greece's Green Upskilling Program or Slovenia's Green Jobs initiative, were introduced to support the green transition. Similarly, Egypt's Takaful and Karama cash transfer programs were launched following energy subsidy reforms. Other programs have extended existing ones; for example, the Philippines expanded its public work program to support fishers affected by a temporary fishing ban in Balayan Bay in 2014. Bangladesh also expanded its Vulnerable Group Feeding program to provide rice to fishermen affected by hilsa fishing closures.

Most country cases are found in high-income countries, in part because they have announced the larger number of green transition policies, particularly decarbonization measures, and many accompanying social protection measures.

The evidence on the impact of different green transition policies on people and workers is still scant. Evidence is emerging on the skills, profile, and preferences of workers affected by policies regulating the coal production phase out in several coal-producing countries. Less is known about the consumption and income losses resulting from restrictions on the utilization of natural resources for small producers, fishers, and farmers. Online job posting data and employer surveys are used to understand what skills are needed to perform green jobs, regardless of how green jobs are defined. Finally, the paper points to knowledge gaps on the effectiveness of specific SPL measures designed to mitigate jobs and income losses, to promote access to green jobs and livelihood, and to energy efficient consumption.

From the implementation perspective, the review points to key factors that are associated with the readiness of SPL systems to ensure and enable a just green transition. Effective institutional coordination and robust data are crucial for ensuring a just green transition. Coordination of ministries of social affairs and labor with ministries of agriculture, education, environment, and energy, is essential to design comprehensive policies that advance the green transition and ensure fairness. Strong coordination is also needed between national and local governments, as the localized nature of some transitioning industries, like coal or fisheries, requires place-based policies. Additionally, robust data and information systems interoperable across key agencies are necessary to identify people vulnerable to the green transition and to measure the income and jobs impacts due to green transition policies, which people and workers will be affected, as the basis to better target green transition policies. Regarding labor market data, while some countries set up observatories to monitor employment shifts in the green economy, many countries need to establish additional LMIS to identify skills and occupations in demand in green industries to better orient education and training systems and ALMPs to better prepare their workers and connect them to green industries.

Targeting is particularly important in social protection measures accompanying natural resource management policies, as support for affected poor and vulnerable households should avoid incentivizing new entrants into resource harvesting (fishing or logging). In other contexts, getting the benefits right may be of paramount importance, as generous benefits can lead to reduced labor force participation. In some contexts, it is also important to ensure that the benefits of the green transition are distributed justly across space and gender, as geographically marginalized groups and women may face unique challenges.

As more countries transition to an economic model not based on fossil fuels and overconsumption of natural resources, social protection will become increasingly important to protect and enable households and workers during the transition. While this paper reviews some examples and highlights characteristics of existing programs, more empirical research is needed to understand the efficacy and evolving characteristics of social protection programs to ensure and enable a just green transition.

References

- Acosta, P., C. Özden, J. Lebow, L. Rodriguez, and E. Dahlgren. 2025. *Global Skill Partnerships for Migration*. Washington, DC: World Bank.
- Alexander, N., L. Li, J. Mondragon, S. Priano, and M. Mendes Tavares. 2024. "The Green Future: Labor Market Implications for Men and Women." IMF Working Paper WP/24/156, International Monetary Fund, Washington, DC.
- Alexandri, E., J. Anton, and R. Lewney. 2024. "The Impact of Climate Change Mitigation Policies on European Labour Markets." *Ecological Economics* 216 (February): 108022.
- Alix-Garcia, J., K. Sims, and P. Yañez-Pagans. 2015. "Only One Tree from Each Seed? Environmental Effectiveness and Poverty Alleviation in Mexico's Payments for Ecosystem Services Program." *American Economic Journal: Economic Policy* 7 (4): 1–40.
- Altenburg, T., C. Fischer, K. Huck, A. Kruip, S. Müller, and S. Sörensen . 2017. *Managing Coastal Ecosystems in the Philippines: What Cash for Work Programmes Can Contribute*. Bonn: German Development Institute (DIE). https://www.idos-research.de/uploads/media/Study_94.pdf.
- Anwar, Rubaba, and Yoonyoung Cho. 2019. "Program Brief Vulnerable Group Feeding Bangladesh." World Bank and Korea-World Bank Partnership Facility (KWPF). <https://documents1.worldbank.org/curated/en/461531552967731486/pdf/135379-BRI-PUBLIC-18-3-2019-10-13-54-ProgrambriefonVGFF.pdf>.
- Arévalo-Sánchez, I., J. Heisey, S. Chaudhary, T. Clay, V. Strokova, P. V. Dutta, and C. Andrews. 2024. *The State of Economic Inclusion Report 2024: Pathways to Scale*. Washington, DC: World Bank.
- Arze del Granado, J., D. Coady, and R. Gillingham. 2010. "The Unequal Benefits of Fuel Subsidies: A Review of Evidence for Developing Countries." IMF Working Paper WP/10/202, International Monetary Fund, Washington, DC.
- Azevedo, D., H. Wolff, and A. Yamazaki. 2023. "Do Carbon Taxes Kill Jobs? Firm-Level Evidence from British Columbia." *Climate Change Economics* 14 (2): 1–46.
- Bakkegaard, R., and S. Wunder. 2014. "Bolsa Floresta, Brazil." In *REDD+ on the Ground: A Case Book of Subnational Initiatives Across the Globe*, edited by E. O. Sills, S. Atmadja, C. de Sassi, A. E. Duchelle, D. L. Kweka, I. A. P. Resosudarmo, and William D. Sunderlin, 51–67. Bogor, Indonesia: Center for International Forestry Research (CIFOR). <https://www2.cifor.org/redd-case-book/case-reports/brazil/bolsa-floresta-brazil/>.
- Behrer, P., and H. Pullabhotla. 2024. "Spilling Over: The Benefits of Public Works Projects for Groundwater in India." Policy Research Working Paper 10886, World Bank, Washington, DC.
- Berryman, A. K., Bücker, J., de Moura, F. S., Barbrook-Johnson, P., Hanusch, M., Mealy, P., Farmer, J. D., & del Rio-Chanona, R. M. (2025). "Skill and spatial mismatches for sustainable development in Brazil". INET Oxford Working Paper 2025-08.

- Bladon, A., G. Greig, Y. Okamura, and G. Nico. 2025. *Blue Social Protection Handbook: Protecting People, Fish and Food (English): Leveraging Social Protection and Jobs Interventions for Sustainable Fisheries*. Washington, DC: World Bank.
- Bladon, A., E. Mohammed, B. Hossain, G. Kibria, L. Ali, and E. J. Milner-Gulland. 2018. "Evaluating the Ecological and Social Targeting of a Compensation Scheme in Bangladesh." *PLoS ONE* 13 (6): e0197809.
- Bladon, A., G. Greig, Y. Okamura. 2022. "Connecting Social Protection and Fisheries Management for Sustainability: A Conceptual Framework." Washington, DC: World Bank.
- Bruegge, C., T. Deryugina, and E. Myers. 2019. "The Distributional Effects of Building Energy Codes." *Journal of the Association of Environmental and Resource Economists* 6 (S1).
- Celasun, O., G. Sher, P. Topalova, and J. Zhou. 2023. "Cars and the Green Transition: Challenges and Opportunities for European Workers." IMF Working Paper WP/23/116, International Monetary Fund, Washington, DC.
- Christiansen Luc, Céline Ferré, Tomasz Gajderowicz, Maddalena Honorati, and Sylwia Wrona. 2022a. *Towards a Just Coal Transition – Labor Market Challenges in Eastern Wielkopolska*. World Bank Group Jobs Working paper n. 71.
- Christiaensen, Luc, Céline Ferré, Tomasz Gajderowicz, Elizabeth Ruppert Bulmer, and Sylwia Wrona. 2022b. *Towards a Just Coal Transition – Labor Market Challenges in Silesia*. World Bank Group Jobs Working Paper N. 70.
- Christiaensen, Luc, Céline Ferré, Tomasz Gajderowicz, Elizabeth Ruppert Bulmer, and Sylwia Wrona. 2022c. *Towards a Just Coal Transition – Labor Market Challenges in Lower Silesia*. World Bank Group Jobs Working Paper N.69.
- Conway, S. 2022. "Cod, Culture, and Loss: Thirty Years of the Cod Moratorium in Newfoundland." <https://activehistory.ca/blog/2022/07/04/cod-culture-and-loss-thirty-years-of-the-cod-moratorium-in-newfoundland/>.
- Cook, Colin. 2020. "Energycarer Support in Scotland." Energy Saving Trust, Blog Post, November 25, 2020. <https://energysavingtrust.org.uk/energycarer-support-scotland/>.
- Costella, C., T. Clay, M. Donoghoe, and L. Giron. 2023. *Pathways to Climate Resilient Economic Inclusion*. Washington, DC: World Bank.
- Costella, C. V., E. S. Shabahat, N. Sadiq, and Y. Okamura. 2024. "Social Protection and Jobs for Climate Change Challenges: Current Practice and Future Opportunities." Discussion Paper 2405, World Bank, Washington, DC.
- Crumpler, K., A. Wybieralska, L. Roffredi, E. Tanganelli, C. Angioni, P. Prospero, V. Umulisa, G. Dahlet, S. Nelson, N. Rai, L. S. Schiettecatte, M. Salvatore, J. Wolf, and M. Bernoux. 2024. *Agrifood Systems in Nationally Determined Contributions: Global Analysis—Key Findings*. Rome: Food and Agriculture Organization.

- Cunningham, Honorati, Granata, and Posadas, forthcoming. Measuring Green Jobs: Choosing the Right Measurement Approach for Each Policy Need. World Bank Group.
- Damania, R., E. Balseca, C. De Fontaubert, J. R. Gill, K. Kim, J. Rentschler, J. Russ, and E. Zaveri. 2023. *Detox Development: Repurposing Environmentally Harmful Subsidies*. Washington, DC: World Bank.
- D’Arcangelo, F. M., I. Levin, A. Pagani, M. Pisu, and Å. Johansson. 2022. “A Framework to Decarbonise the Economy.” OECD Economic Policy Paper 31, Organisation for Economic Co-operation and Development, Paris.
https://www.oecd.org/content/dam/oecd/en/publications/reports/2022/02/a-framework-to-decarbonise-the-economy_642a6827/4e4d973d-en.pdf.
- Deininger, K., and Y. Liu. 2019. “Heterogeneous Welfare Impacts of National Rural Employment Guarantee Scheme: Evidence from Andhra Pradesh, India.” *World Development* 117 (May): 98–111.
- Dias, S. 2016. “Waste Pickers and Cities.” *Environment and Urbanization* 28 (2): 375–90.
- Données et études statistiques. 2023. “Emplois et métiers de l’économie verte—Synthèse des connaissances en 2023.” <https://www.statistiques.developpement-durable.gouv.fr/emplois-et-metiers-de-leconomie-verte-synthese-des-connaissances-en-2023?rubrique=40&dossier=190>.
- Dorband, I. I., M. Jakob, M. Kalkuhl, and J. C. Steckel. 2019. “Poverty and Distributional Effects of Carbon Pricing in Low- and Middle-Income Countries—A Global Comparative Analysis.” *World Development* 115 (March): 246–57.
<http://sciencedirect.com/science/article/pii/S0305750X18304212>.
- Dubey, S., E. Adovor, D. Rysankova, E. Portale, and B. Koo. 2019. *Kenya: Beyond Connections Access to Electricity and Clean Cooking in Kenya Based on the Multitier Framework Survey and Data Analysis*. Energy Sector Management Assistance Program (ESMAP) report. Washington, DC: World Bank.
- European Commission. 2022. Fairness Perceptions of the Green Transition. Special Eurobarometer 527.
- European Commission. 2023a. “PES Support to Greening of the Labour Market.” European Commission, Brussels. <https://op.europa.eu/en/publication-detail/-/publication/04ec652e-1ae8-11ee-806b-01aa75ed71a1>.
- European Commission. 2023b. “State Aid: Commission Approves Amendments to Polish Scheme to Support Closure of Coal Mines.” European Commission, Brussels.
https://energy.ec.europa.eu/news/state-aid-commission-approves-amendments-polish-scheme-support-closure-coal-mines-2023-05-05_en.
- European Commission. N.d. “The Green Deal.” European Commission, Brussels.
https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en.
- European Training Foundation. 2022. “Skills for the Green Transition.”
<https://www.etf.europa.eu/sites/default/files/2022->

11/Edited%20green%20transition%20policy%20brief_EN.pdf?trk=public_post_comment-text#:~:text=The%20green%20transition%20is%20a%20process%20towards%20a,atmosphere%20%20as%20well%20as%20the%20loss%20of%20bio.

FAO (Food and Agriculture Organization). 2022. "Impact Evaluation of the Fishers' Unemployment Insurance in Brazil." Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries. <https://www.fao.org/voluntary-guidelines-small-scale-fisheries/news/news-detail/Impact-evaluation-of-the-fishers-unemployment-insurance-in-Brazil/en>.

FAO. 2023. *Strengthening Coherence Between Forestry and Social Protection for Sustainable Agrifood Systems Transformation—Framework for Analysis and Action*. Rome: Food and Agriculture Organization. doi:<https://doi.org/10.4060/cc8648en>.

Ferraro, P., and R. Simorangkir. 2020. "Conditional Cash Transfers to Alleviate Poverty Also Reduced Deforestation in Indonesia." *Science Advances* 6 (24): eaaz1298.

Forestry Commission. 2021. *Ghana Forest Plantation Strategy: Annual Report 2020*. Ghana: Forestry Commission. <https://fcghana.org/wp-content/uploads/2021/12/null1.pdf>.

Galgóczi, B. 2019. "Phasing Out Coal—A Just Transition Approach." ETUI Working Paper 2019.04, European Trade Union Institute, Brussels.

Gasior, K., G. Wright, D. McLennan, M. Noble, H. X. Jara, D. Malerba, M-C. Dankmeyer, and S. Beierl. 2024. *The Role of Social Protection for a Just Transition in Developing and Emerging Economies*. Bonn: German Agency for International Cooperation (GIZ).

GCF (Governors' Climate and Forests) Task Force. 2022. "Launch of the 'Guardians of the Forest' and the Manaus Action Plan Marks the Opening of the 12th Annual Meeting of the GCF in Manaus." GCF Task Force, March 17, 2022. <https://www.gcftf.org/launch-of-the-guardians-of-the-forest-and-the-manaus-action-plan-marks-the-opening-of-the-12th-annual-meeting-of-the-gcf-in-manaus/>.

Gencer, D., and B. Arizu. 2024. *From Ambition to Action: Practical Insights on Energy Subsidy Reforms*. Energy Sector Management Assistance Program (ESMAP) Report. Energy Subsidy Reform in Action Series. Washington, DC: World Bank. <https://openknowledge.worldbank.org/entities/publication/f96a67e1-dcb8-40f7-90ed-9622f6e692f5>.

Ghana Climate Innovation Centre. N.d. "Enabling Africa's Climate Resilience." <https://ghanacic.ashesi.edu.gh/>.

Ghana Youth Employment Agency. N.d. "Agenda for Jobs." , <https://yea.gov.gh/home/>

Ghose, D., E. Fraga, and A. Fernandes. 2023. "Fertilizer Import Bans, Agricultural Exports, and Welfare: Evidence from Sri Lanka." Policy Research Working Paper 10642, World Bank, Washington, DC.

Godinho, C. 2022. "What Do We Know About the Employment Impacts of Climate Policies? A Review of the Ex Post Literature." *Wiley Interdisciplinary Reviews: Climate Change* 13 (6): e794.

- Government of Antigua and Barbuda. 2021. "Antigua and Barbuda Updated Nationally Determined Contribution: For the Period 2020–2030." <https://unfccc.int/sites/default/files/NDC/2022-06/ATG - UNFCCC NDC - 2021-09-02 - Final.pdf>.
- Government of Canada. 2025. "Removing the Consumer Carbon Price, Effective April 1, 2025." <https://www.canada.ca/en/departement-finance/news/2025/03/removing-the-consumer-carbon-price-effective-april-1-2025.html>.
- Government of Newfoundland and Labrador. N.d. "Major Milestones." <https://www.gov.nl.ca/publicat/tags/text/page34.htm>.
- Granata, J., and J. Posadas. 2024. "Why Look at Tasks When Designing Skills Policy for the Green Transition? A Methodological Note on How to Identify Green Occupations and the Skills They Require." Policy Research Working Paper 10753, World Bank, Washington, DC.
- Green Climate Fund. 2018. "FP062; Poverty, Reforestation, Energy and Climate Change Project (PROEZA)." <https://www.greenclimate.fund/project/fp062>.
- Green Climate Fund. 2024. *Poverty, Reforestation, Energy and Climate Change Project (PROEZA) Annual Performance Report CY2023*. Incheon, Republic of Korea: Green Climate Fund. <https://www.greenclimate.fund/sites/default/files/document/fp062-annual-performance-report-cy2023.pdf>.
- Györi, M., K. Diekmann, and E. Kühne. 2021. *The Importance of Social Protection for Climate Change Mitigation in LMICs: Success Stories and Opportunities for the Future*. Bonn: German Agency for International Cooperation (GIZ).
- Hafstead Marc A.C. and R. C. Williams. 2020. "Jobs and Environmental Regulation," *Environmental and Energy Policy and the Economy*, University of Chicago Press, vol. 1(1), pages 192-240
- Hallegatte, S., C. Godinho, J. Rentschler, P. Avner, I. Dorband, C. Knudsen, J. Lemke, and P. Mealy. 2024. *Within Reach: Navigating the Political Economy of Decarbonization*. Washington, DC: World Bank.
- Ham, A., E. Vazquez, and M. Yanez-Pagans. 2025. "Characterizing Green and Carbon-Intensive Employment in India." *Ecological Economics* 236 (October): 108695.
- Ham, A., E. Vazquez, M. Yanez-Pegans, C. Knudsen, and S. Asad. Forthcoming. *Occupational Mobility and the Green Transition: Estimating Skill Investment Needs in South Asia*.
- Haque, A. B. M. M. 2020. "Hilsa Production and Fishers' Income Increased Due to Co-Management Strategies Aimed to Enhance the Socio-Economic Resilience of Fishing Communities in Bangladesh." WorldFish, Penang, Malaysia. <https://mel.cgiar.org/projects/ecofish/214/hilsa-production-and-fishers-income-increased-due-to-co-management-strategies-aimed-to-enhance-the-socio-economic-resilience-of-fishing-communities>.
- Higgins, J. 2008. "Economic Impacts of the Cod Moratorium." <https://www.heritage.nf.ca/articles/economy/moratorium-impacts.php#:~:text=Known%20as%20the%20Northern%20Cod,%24225%20to%20%24406%20a%20week>.

- Hiroven, K., E. Machado, A. Simons, and V. Taraz. 2022. "More Than a Safety Net: Ethiopia's Flagship Public Works Program Increases Tree Cover." *Global Environmental Change* 75 (July): 102549.
- Honorati, M., and A. Banaszczyk. 2022. "Options to Support Workers Through a Transition Away from Coal in Eastern Wielkopolska." Jobs Notes Issue No.16, World Bank, Washington, DC. <https://openknowledge.worldbank.org/server/api/core/bitstreams/0b65a953-5293-4f85-ae79-aa87a562e96a/content>.
- Huckstep, S., and H. Dempster. 2024. "Meeting Skill Needs for the Global Green Transition: A Role for Labour Migration?" CGD Policy Paper 318, Center for Global Development, Washington, DC. <https://www.cgdev.org/sites/default/files/meeting-skill-needs-global-green-transition-role-labour-migration.pdf>.
- IEA (International Environment Agency). 2022a. *World Energy Outlook 2022*
- IEA. 2022b. *Skills Development and Inclusivity for Clean Energy Transitions*.
- ILO (International Labour Organization). 2015a. "Guidelines for a Just Transition Towards Environmentally Sustainable Economies and Societies for All." https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40ed_emp/%40emp_ent/documents/publication/wcms_432859.pdf.
- ILO (International Labour Organization). 2015b. *Just Transition for All: Can the Past Inform the Future?* International Journal of Labour Research 6 (2): 173-185.
- ILO (International Labour Organization). 2018. *Greening with Jobs – World Employment and Social Outlook 2018*. Geneva: International Labor Organization.
- ILO. 2022. "Social Protection in Action: Building Social Protection Floors for All. China: Social Protection and Climate Change." <https://www.social-protection.org/gimi/Media.action?id=19386>.
- ILO. 2024. *World Social Protection Report 2024-26: Universal Social Protection for Climate Action and a Just Transition*, Geneva: International Labor Office, 2024.
- IRENA and ILO. 2023. Renewable energy and jobs: Annual review 2023, International Renewable Energy Agency, Abu Dhabi and International Labor Organization, Geneva.
- Islam, M., E. Y. Mohammed, and L. Ali. 2014. "Characteristics and History of Hilsa Shad Fishery." In *Economic Incentives for Sustainable Hilsa Fish Management in Bangladesh: An Analysis of the Legal and Institutional Framework*, 11–14. International Institute for Environment and Development. <http://www.jstor.org/stable/resrep18083.7>.
- Islam, M., E. Y. Mohammed, and L. Ali. 2016. "Economic Incentives for Sustainable Hilsa Fishing in Bangladesh: An Analysis of the Legal and Institutional Framework." *Marine Policy* 68 (June): 8–22.
- Jobs and Skills Australia. 2023. *The Clean Energy Generation: Workforce Needs for a Net Zero Economy*. Canberra: Jobs and Skills Australia. <https://www.jobsandskills.gov.au/download/19313/clean-energy-generation/2385/clean-energy-generation/pdf>.

- Kagombe, J., J. Kiprop, D. Langat, and J. Cheboiwo. 2020. *Socio-Economic Impacts of Forest Harvesting Moratorium in Kenya*. Technical Report. Nairobi, Kenya: Kenya Forestry Research Institute.
https://www.researchgate.net/publication/343549071_TECHNICAL_REPORT_SOCIO-ECONOMIC_IMPACT_OF_FOREST_HARVESTING_MORATORIUM_IN_KENYA.
- Keese, M., and L. Marcolin. 2023. "Labour and Social Policies for the Green Transition: A Conceptual Framework." OECD Social, Employment and Migration Working Paper 295, Organisation for Economic Co-operation and Development, Paris.
- Knudsen, C., F. Senra de Moura, J. Bücker, and P. Mealy. Forthcoming. *Five Frictions: Key Labor Market Barriers to Unlocking Job Growth in the Green Transition*. Washington, DC: World Bank.
- Levinson, A. 2019. "Energy Efficiency Standards Are More Regressive Than Energy Taxes: Theory and Evidence." *Journal of the Association of Environmental and Resource Economists* 6 (51): S7–S36.
- Malerba, D. 2020. "Poverty Alleviation and Local Environmental Degradation: An Empirical Analysis in Colombia." *World Development* 127 (March): 104776.
- May, J., M. Velia, and O. Benammour. 2023. *The Role of Social Protection in Strengthening Agrifood Systems and Inclusive Rural Transformation*. Rome: FAO.
<https://openknowledge.fao.org/server/api/core/bitstreams/42898d3e-9110-459e-bfee-3462b7441832/content>.
- Mayer Gukovas, R., D. Garrote-Sanchez, and M. Makovec. Forthcoming. "Who's at Risk in the Green Transition? An Analysis of the Extensive and Intensive Margin Across Countries in Europe and Central Asia."
- Meioambiente. 2024. "Social Security Inclusion Project for Recyclable Material Collectors in Minas Gerais Will Be Launched This Tuesday." Minas Gerais Meioambiente.
<https://meioambiente.mg.gov.br/w/projeto-de-inclusao-previdenciaria-de-catadores-de-materiais-reciclaveis-em-minas-sera-lancado-nesta-terca>.
- Ministry of Environment and Climate Change. N.d. "Bolsa Verde." Ministry of Environment and Climate Change, Government of Brazil. <https://www.gov.br/mma/pt-br/composicao/snpct/dpct/bolsa-verde>.
- Ministry of the Environment and Sustainable Development. 2023. *Emission Reduction Program Around Tai National Park: Benefit Sharing Plan*. Yamoussoukro, Côte d'Ivoire: Ministry of the Environment and Sustainable Development, Republic of Côte d'Ivoire.
https://www.forestcarbonpartnership.org/sites/default/files/documents/cote_d_ivoire_final_benefit_sharing_plan_updated_december_12_2023.pdf.
- Ministry of Petroleum and Natural Gas. *Report of the Comptroller and Auditor General of India on Pradhan Mantri Ujjwala Yojana*. Report No. 14 of 2019.
https://www.cag.gov.in/uploads/download_audit_report/2019/Report_No_14_of_2019_Performance_Audit_of_Pradhan_Mantri_Ujjwala_Yojana_Ministry_of_Petroleum_and_Natural_Gas_0.pdf

- Mosomi, J. C., and W. Cunningham 2024. "Profiling Green Jobs and Workers in South Africa: An Occupational Tasks Approach." SALDRU Working Paper Number 305, Southern Africa Labour and Development Research Unit, University of Cape Town, Cape Town.
- Mukherjee, A., Y. Okamura, U. Gentilini, D. Gencer, M. B. Almenfi, A. Kryeziu, and N. Umapathi. 2023. *Energy Subsidy Reform in Action: Cash Transfers in the Context of Energy Subsidy Reform—Insights from Recent Experience*. Technical Report, Energy Subsidy Reform in Action series. Washington, DC: World Bank.
- Nandwani, B., and M. Jain. 2024. "Access to Clean Cooking Fuel and Women Outcomes." Working Paper WP-2024-017, Indira Gandhi Institute of Development Research, Mumbai.
<http://www.igidr.ac.in/pdf/publication/WP-2024-017.pdf>.
- NDC Partnership. 2025. "Placing People at the Centre of NDCs 3.0: The Power of Social Protection to Accelerate Climate Action." <https://ndcpartnership.org/sites/default/files/2025-02/guidance-noteplacing-people-centre-ndcs-30.pdf>.
- OECD (Organisation for Economic Co-operation and Development). 2022. OECD Environment Working Papers 203.
- OECD. 2023. Job Creation and Local Economic Development 2023: Bridging the Great Green Divide, OECD Publishing, Paris, <https://doi.org/10.1787/21db61c1-en>.
- Olivier, A., M. Matytsin, and G. Gencer. 2023. *Energy Subsidy in Action: Distributional Analysis for Informing Energy Subsidy Reforms—Review of Recent Approaches*. Energy Sector Management Assistance Program (ESMAP) Technical Report. Washington, DC: World Bank.
<https://openknowledge.worldbank.org/entities/publication/cb894829-90f3-42af-b2f1-33453f8ce106>.
- OXFAM. 2020. "Protecting People and the Forest, Bolsa Verde, Brazil." <https://policy-practice.oxfam.org/resources/protecting-the-people-and-the-forest-bolsa-verde-brazil-621099/>.
- Testaverde, M. , J. Posadas, I. Cho, M.J. Granata, S. Johansson De Silva, D. G. Kim, D. Gyunam, I. Meidina, C.T. Seooun, P. Prakosa, B. Wiyaktiand K. Yoo. 2021. *Toward a World-Class Labor Market Information System for Indonesia : An Assessment of the System Managed by (English)*. Washington, D.C. : World Bank Group.
- Presidency Republic of South Africa, The. 2023. *Just Energy Transition Implementation Plan 2023–2027*. Pretoria: Presidency Republic of South Africa.
<https://www.stateofthenation.gov.za/assets/downloads/JET Implementation Plan 2023-2027.pdf>.
- Presidency Republic of South Africa, The. 2025. *Just Energy Transition Implementation Plan (JET IP): Quarterly Progress Report, 31 March 2025*. Pretoria: The Presidency Republic of South Africa.
<https://justenergytransition.co.za/wp-content/uploads/2025/04/JetQuarterly31March2025.pdf>.
- Press Information Bureau. 2022. "Direct Benefit Transfer (DBT): Transforming Governance in Uttar Pradesh: Nearly Rs 76000 Crore Transferred to Beneficiaries Under DBT in the State."
<https://static.pib.gov.in/WriteReadData/specificdocs/documents/2022/jan/doc2022153101.pdf>.

- RENEW MENA (Regional Network in Energy for Women in the Middle East and North Africa). 2024. *Annual Report 2023*. Washington, DC: World Bank.
https://collaboration.worldbank.org/content/sites/collaboration-for-development/en/groups/renew-mena/documents.entry.html/2025/04/07/renew_mena_annualreport2024-Xr41.html.
- Republic of Kenya, Ministry of Energy and Petroleum. 2024. *Kenya National Cooking Transition Strategy 2024–2028*. Nairobi: Republic of Kenya, Ministry of Energy and Petroleum.
- Republic of Malawi. 2021. “Updated Nationally Determined Contributions.”
[https://unfccc.int/sites/default/files/NDC/2022-06/Malawi Updated NDC July 2021 submitted.pdf](https://unfccc.int/sites/default/files/NDC/2022-06/Malawi%20Updated%20NDC%20July%202021%20submitted.pdf).
- Rigolini, J. 2021. “Social Protection and Labor: A Key Enabler for Climate Change Adaptation and Mitigation.” Social Protection and Jobs Discussion Paper 2108, World Bank, Washington, DC.
- Ruppert Bulmer, E., K. Pela, A. Eberhard-Ruiz, and J. Montoya. 2021. *Global Perspective on Coal Jobs and Managing Labor Transition out of Coal: Key Issues and Policy Responses*. Washington, DC: World Bank.
- Rutkowski J. , C. de Paz and V. Levin. 2018. "Labor Market Observatories: critical success factors." World Bank Jobs Notes Issue n. 4.
- Sabarwal, S., S. Venegas Marin, M. H. Spivack, and D. Ambasz. 2024. *Choosing Our Future-Education for Climate Action*. Washington, DC: World Bank.
- Saget, C., A. Vogt-Schilb, and T. Luu. 2020. *Jobs in a Net-Zero Emissions Future in Latin America and the Caribbean*. Washington, DC and Geneva: Inter-American Development Bank and International Labour Organization.
- SantaFe Provincia. 2018. “The Province Is Committed to Generating Green Jobs.”
<https://www.santafe.gov.ar/noticias/noticia/262850/>.
- SCGJ (Skill Council for Green Jobs). 2024. *Annual Report 2023–24*. New Delhi, India: SCGJ.
<https://sscgj.in/wp-content/uploads/2024/10/Annual-Report-2023-24.pdf>.
- Tomic, I., A. Sitarova, K. S. Pela, A. Budimir, L. K. Matić, and M. Testaverde. 2025. *Empowering Croatia's Future Enhancing Youth Employment and Skills Development*. Washington, DC: World Bank.
- UNFCCC (United Nations Framework Convention on Climate Change). 2023. “Leaving No One Behind in the Transition Towards a Low-Carbon Economy.” UN Climate Change News, April 12, 2023.
<https://unfccc.int/news/leaving-no-one-behind-in-the-transition-towards-a-low-carbon-economy>.
- Vona, F., G. Marin, D. Consoli, and D. Popp. 2018. “Environmental Regulation and Green Skills: An Empirical Exploration.” *Journal of the Association of Environmental and Resource Economists* 5 (4): 713–53.
- Winkler, H., V. Di Maro, K. Montoya Munoz, S. Olivieri, and E. Vazquez. 2024. “Measuring Green Jobs: A New Database for Latin America and Other Regions.” Policy Research Working Paper 10794, World Bank, Washington, DC.

- Wong, P., K. Kuralbayeva, L. Anderson, A. M. Pessoa, and T. Harding. 2025. "Individual Pay for Collective Performance and Deforestation: Evidence from Brazil." <https://sites.google.com/site/poyinpywong/research>.
- World Bank 2011. "Romania Poverty and Social Impact Assessment." **[[provide URL]]**
- World Bank. 2012. *Inclusive Green Growth: The Pathway to Sustainable Development*. Washington, DC: World Bank. <https://openknowledge.worldbank.org/server/api/core/bitstreams/8fd533cc-8dd2-5a0a-820b-2de519108f52/content>.
- World Bank. 2016. "Climate Innovation Center Launched to Support Ghana's Green Economy." Press release, May 17, 2016. <https://www.worldbank.org/en/news/press-release/2016/05/17/climate-innovation-center-launched-to-support-ghanas-green-economy>.
- World Bank. 2018. *Managing Close Mine Closures: Achieving a Just Transition for All*. Washington, DC: World Bank.
- World Bank. 2021a. *Summary Report: Identifying Capacity Building Needs for Carbon Pricing and International Carbon Market Development Through the Mitigation Action Assessment Protocol (MAAP)*. Washington, DC: World Bank.
- World Bank. 2022a. *Toward More and Better Jobs for Women in Energy*. Washington, DC: World Bank.
- World Bank. 2022b. *Charting a Course Towards Universal Social Protection: Resilience, Equity, and Opportunity for All*. Washington, DC: World Bank.
- World Bank. 2022c. *South Africa - Country Climate and Development Report*. Washington, DC: World Bank.
- World Bank. 2023a. *PAD for China Green Agricultural and Rural Revitalization Program for Results (Hubei and Hunan)*. Washington, DC: World Bank.
- World Bank. 2023b. *Reality Check: Lessons from 25 Policies Advancing a Low-Carbon Future*. Climate Change and Development Series. Washington, DC: World Bank.
- World Bank. 2024a. "In Brazil, Forest Entrepreneurs Showcase the Vast Potential of the Amazon to the Entire World." Feature story, World Bank, Washington, DC, July 1, 2024. <https://www.worldbank.org/en/news/feature/2024/07/01/brazil-forest-entrepreneurs-showcase-potential-amazon>.
- World Bank. 2024b. *People in a Changing Climate: From Vulnerability to Action*. Country Climate and Development Report Synthesis. Washington, DC: World Bank.
- World Bank. 2024c. *Implementation Completion and Results Report for the Mongolia Ulaanbaatar Clean Air Project (P122320)*. Washington, DC: World Bank. <https://documents1.worldbank.org/curated/en/099062024100569664/pdf/BOSIB14333ddfe0cc1a34e1f863a3c3db67.pdf>.
- World Bank. 2024d. *Poland Country Climate and Development Report*. Washington, DC: World Bank.

- World Bank. 2025a. *Western Balkans Regular Economic Report No. 27: Adapting for Sustainable Growth*. Washington, DC: World Bank.
- World Bank. 2025b “Labor Market Challenges and Policy Responses in China’s Green and Low-Carbon Transition.” Shanxi case study report (mimeo), World Bank, Washington, DC.
- World Bank. Forthcoming. *Innovations in Public Works*. Washington, DC: World Bank.
- WWP (World Without Poverty). 2017. “Environmental Conservation Support Program: Bolsa Verde (Green Grant).” <https://wwp.org.br/wp-content/uploads/02.-Bolsa-Verde-Program-Sheet.pdf>.
- Yadav, Yashi. 2020. “A Socio-Economic Assessment of Pradhan Mantri Ujjwala Yojana.” *International Journal of Humanities and Social Science Invention (IJHSSI)* 9 (1): 26–30.
[https://www.ijhssi.org/papers/vol9\(1\)/Series-1/F0901012630.pdf#:~:text=the%20government%20to%20provide%20cooking,The.](https://www.ijhssi.org/papers/vol9(1)/Series-1/F0901012630.pdf#:~:text=the%20government%20to%20provide%20cooking,The.)
- Yamazaki, A. 2017. “Jobs and Climate Policy: Evidence from British Columbia's Revenue-Neutral Carbon Tax.” *Journal of Environmental Economics and Management* 83 (May): 197–216.
- Yemtsov, R., and A. Moubarak. 2018. “Readiness Assessment of Social Safety Nets to Support Energy Subsidy Reform.” ESMAP Papers, World Bank, Washington, DC.
<https://openknowledge.worldbank.org/entities/publication/2d0f1ce1-0a24-5402-a484-65daf57298c3/full>.
- Yip, C. M. 2018. “On the Labor Market Consequences of Environmental Taxes.” *Journal of Environmental Economics and Management* 89 (May): 136–52 .
- Yuen, Simon. 2023. “BSW Inks Deal to Integrate Indian Workers into German Solar Industry.” PV Tech, February 27, 2023. <https://www.pv-tech.org/bsw-inks-deal-to-integrate-indian-workers-into-german-solar-industry/>.
- Zhang, Y., and N. Adams. 2016. “Toward Universal Access to Clean Cooking and Heating: Early Lessons from the East Asia and Pacific Clean Stove Initiative.” Live Wire 2016/62, World Bank, Washington, DC.

Annex 1. Overview of selected policy databases

International databases (with hyperlinks)	Policies covered	Number of policies	Period spanned	Number of countries	Policy instrument categories	Comments
OECD's Policy Instruments for the Environment (PINE)	Environmental policies	4,590	1900–2024 (median policy 2002)	151	5 types; 19 subtypes	Very rich—over 400 variables per policy
OECD's Climate Actions and Policies Measurement Framework (CAPMF)	Climate policies	at least ~1,500	1990–2022	52	3 categories; 11 types; 56 instruments	Rich, but focused on covering a quantitative measure of stringency (or existence) rather than the detailed qualitative policy descriptions offered in other databases.
World Bank's Carbon Policies Database	Carbon pricing policies	181	1990–2024	106 with ~50 subnational (counting “compliance” type)	2 major categories “compliance” (tax and ETS) and “crediting”	Comprehensive coverage of carbon pricing (including prices and share of emissions covered).
IEA Policies Database	Energy policies and measures to reduce greenhouse gas emissions	8,065	1948–2025 (median 2013)	163	190 policy types	Detailed qualitative descriptions of policies sourced with a link to reference information—focus on energy policies.
Climate Policies Database	Climate policies	6,273	1979–2026 (median 2018, incl. planned only)	198	9 categories, 8 further subcategories, 54 policy instruments	Detailed qualitative descriptions with sourcing—focus on climate policies.
Climate Policy Radar	Climate policy documents (laws and regulations)	6,991	1947–2024 (median 2018)	202	5 instruments, 26 subvalues	Heavy focus on using artificial intelligence to search, categorize, and improve database.

Source: Dashboard of green transition policies.

Note: ETS = emissions trading system

Annex 2. OECD climate actions and policies measurement framework

Instrument category	Instrument type	Instruments (selected examples)
Market-based instruments	Taxes and fees	Carbon taxes Fuel excise taxes Congestion charges
	Trading systems	Emission trading systems (including CORSIA) Tradable renewable energy certificates
	Subsidies	Feed-in-tariffs and auctions for renewable electricity Research Demonstration and Development for low-carbon energy Financing mechanisms for energy efficiency (eg preferential loans)
Non market-based instruments	Performance standards	Air emissions standards for coal power plants Minimum energy performance standards for electric motors and vehicles Building codes
	Technology standards	Bans and phase-outs of fossil fuel-based assets and equipment Bans on financing fossil fuel-based infrastructure abroad
	Information instruments	Energy labels for appliances Energy labels for passenger cars
	Other non market-based instruments	Public investment in rail infrastructure Policies to reduce fugitive methane emissions Planning instruments for renewables expansion
Targets, international co-ordination, climate governance and climate data	Targets	Nationally Determined Contributions (NDCs) Net-zero targets
	International co-ordination	Participation in international climate initiatives Ratification of key climate treaties
	Climate governance	Climate advisory bodies
	Climate data	GHG emissions reporting and accounting Evaluation of Biennial (Update) Reports Submissions to UNFCCC

Source: OECD 2022.

Note: CORSIA = Carbon Offsetting and Reduction Scheme for International Aviation; GHG = greenhouse gas; UNFCCC = United Nations Framework Convention on Climate Change.

Social Protection & Jobs Discussion Paper Series Titles FY 2026

<u>No.</u>	<u>Title</u>
2602	Supporting People and Workers in an Inclusive Green Transition: Framing Social Protection and Labor Policies to Enable and Ensure a Just Green Transition
2601	Refugees and social assistance in low- and middle-income countries: A review of operational experiences

To view Social Protection & Jobs Discussion Papers published prior to 2021, please visit www.worldbank.org/sp.

ABSTRACT

This paper discusses the role of social protection and labor (SPL) policies in complementing green transition policies by both mitigating the potential negative impacts of the transition on workers and low-income households and by enabling people to take advantage of emerging opportunities. The paper proposes a conceptual framework to assess how green transition policies influence people through changes in prices, production, and jobs. The paper reviews country case studies of SPL programs designed to protect people exposed to the potential negative impacts of the green transition, to prepare workers for green jobs and to promote environmentally sustainable livelihoods. Key factors of success in integrating SPL policies into green transition policies include effective institutional coordination, strong delivery systems, and integrated data systems.

JEL CODES

I38, J68, H53, Q58.

KEYWORDS

Green transition, just transition, social protection, labor, climate mitigation.

